

DENTOBOT

Step 6 Planner Recovery & Simulation Studio Foundation

A-Z implementation, evidence and decision report

S6-LIVE-01 | FDI21 / FDI31 | 23 September 2026

Prepared for Tarun to review before selecting the next development and runtime action.

CURRENT DISPOSITION: runtime paused; no completed two-target, three-planner comparison; native FDI21 Stage-3 collision retained; offline collision-image attribution defect identified and not yet repaired.

Scope: the supplied conversation, development-controlled records, current source checkpoint, retained run artifacts and available screenshots. This is a decision-support report, not a new execution contract or permission to run simulation.

Research simulation only. No hardware motion, drilling authorization, clinical validation or calibrated physical-fit claim.

Repository: /home/light-tarun/dentobot/ros2_ws/src/DentoBot

Branch: cursor-agent/dentoworkflow-debug-20260918

HEAD at report preparation: 48ad8a8e488f396e52f8ed8bbaba679e2e6b4e99

Working tree: dirty, with comparison and diagnostic changes not represented by HEAD alone.

Abstract

The original problem was a collision-aware Home-to-PreEntry connectivity failure in an FDI31 case. The discussion subsequently exposed several different problems: diagnostic-URDF launch and profile compatibility, persistent per-target guide lineage, stale motion evidence after switching targets, the distinction between IK reachability and collision-valid motion, and the cost and observability of a three-planner comparison. They are related prerequisites, but they are not one planner defect.

Operator trials demonstrated that an altered FDI31 setup could complete all three planning stages with spindle collision enabled after changes to mouth opening, Task Home and robot-base placement/orientation. That is meaningful feasibility evidence for that particular setup. It does not identify which change was causal, validate another target, or establish a reproducible canonical baseline without the exact adjusted case.

The latest timed FDI21 run completed one RRTConnect trial: Stage 1 and Stage 2 passed, but the independent Stage-3 guard rejected target-tooth versus pneumatic-spindle contact. RRT began but was interrupted by the 20-minute batch cap; RRT* had no completed result. Approximately 813 seconds across eight waypoint-guard calls dominated the approximately 874-second RRTConnect trial. This is not a fair comparison of the three planners.

The attempted offline contact pictures were rejected by the operator because they did not show the reported collision. A source audit then found that the recapture script applied the lower-jaw opening transform to the fixed upper FDI21 tooth. Correcting that evidence path is the immediate bounded implementation step. It is not yet a fix for native collision or planner feasibility.

The recommended decision path is finite and causal: repair evidence attribution; reconcile one exact rejected state; determine which geometric/kinematic constraint blocks insertion; consider only mechanism-backed base/Home changes within declared constraints; then optimize measured repeated guard work and complete the frozen-input comparison. Manual tuning until green is rejected as the main development strategy. A useful Studio must also report credible failure or uncertainty, not manufacture universal success.

How to read this report

Read Sections 1-5 for the current status and history, Sections 6-12 for technical diagnosis and options, and Sections 13-16 for the implementation boundary, decision gates and evidence references. The screenshot atlas follows, with a separate caption and limitation for each image. A final source appendix preserves detailed run chronology from the logbook.

Evidence labels

OPERATOR OBSERVATION: what Tarun reported or visibly demonstrated. SOURCE FINDING: behavior identified in code or serialized records. RUNTIME EVIDENCE: a recorded run outcome, bound to its inputs. SOURCE-COMPLETE: implemented and checked in isolation, not accepted end to end. HYPOTHESIS: a plausible explanation or adjustment not yet demonstrated. PROPOSED: a future action, not implemented or newly authorized.

Completeness and limitations

This is an exhaustive thematic account of the supplied discussion and available relevant records, not a certified verbatim export of every past assistant response. Some earlier turns are represented by preserved logbook entries and conversation continuity rather than complete raw transcripts. Undated chat messages are sequenced, not assigned invented times. Missing results remain missing; screenshots are not treated as proof of unseen steps.

Document authority

The canonical execution contract remains docs/diagnostics/FDI31_GUI_PLANNER_PO_PLAN_2026-09-21.md. This report is a dated evidence snapshot. It creates no parallel pending queue and grants no new runtime or geometry-change authority. Later task-specific operator decisions supersede older generic paragraphs in the records. Engineer-owned journals and trackers were not read.

Reading map / contents

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1. Outcome sought and present position

Operator objective

Prepare multiple target-tooth trajectories and respective guides in one subject case; switch between completed guides without unrelated staleness; verify the selected guide at Step 5C; activate its PreparedBranch; run Step 6 simulation and retain all diagnostic evidence. Ultimately a non-engineer dentist should be able to operate a robust multi-trajectory Simulation Studio, with the engineering complexity handled by the product.

Immediate comparison contract

One Compare Three Planners action runs configured RRTConnect, RRT and RRT* sequentially against identical target, base, Task Home, acknowledged scene and settings. Default settings are one attempt and 5.0 seconds per native joint-planning request. Ordinary planning or guard failures are retained; fatal runtime/identity failures and cancellation leave explicit NotRun rows. No automatic winner or motion authority is created.

Where we actually stand

The comparison action, branch-keyed persistence, report UI and display-only replay have a recorded source implementation and focused checks. Runtime acceptance is incomplete. The latest FDI21 record is a checkpoint, not a completed saved comparison case. FDI31 continuation and ROS-disconnected reopen acceptance of a completed comparison have not followed. The screenshot attribution defect is the immediate blocker to informed human geometry review.

What a successful foundation means

Success is not that all 96 registry slots or roughly 100 proposed trajectories must pass. Success means a reproducible, auditable system that distinguishes feasible tasks, demonstrated constraints, not-yet-demonstrated feasibility and inconclusive input/evidence. It must make adjustment advice understandable without hiding failed attempts or altering the task silently.

2. Status at a glance

Area	Current evidence and remaining boundary
FDI31 canonical recovery	Historical P1 failure; diagnostic housing-off P1 pass; operator alternate-base housing-on full-chain pass. Exact altered working scene not independently reproduced; canonical milestone closure remains open.
FDI21 guide / branch setup	Two-target regeneration, alternating selection and fresh reopen have recorded headless successes. User reported working verification; broad normal-window acceptance must not be inferred from unrelated planner screenshots.
Planner controls	RRTConnect, RRT and RRT* configured; attempts/time controls; tooltips and reopenable error text. Operator confirmed tips and retained message. Approximate IK locked off; Cartesian Stage 2/3 still locked on.
Three-planner batch	Source implemented and focused checks recorded. Full three-row meaningful runtime comparison for FDI21/FDI31 is not complete.
Latest FDI21 r6	RRTConnect Stage 1/2 pass, Stage 3 native collision; RRT interrupted; RRT* no completed result; overall timeout.
Collision pictures	Three offline capture attempts; final images rejected. Fixed-upper tooth incorrectly jaw-transformed by recapture script. Source-only bounds check exists; repair and corrected runtime evidence pending.
Time cost	Waypoint guard dominates measured first trial. Increasing timeout adds capacity, not geometric feasibility or algorithm correctness.
Studio / full cycle	Scheduler not built in this pass. Guarded Drill, Return Home, repeatability and complete saved replay acceptance remain separate gates.

3. Chronology I - from FDI31 recovery to geometry

Sep-19 baseline and Experiment A

The starting branch/revision supplied by the operator was b79161d93fd37a2ba2fee67703a8229cdf43689b. The sole original runtime case was dentobot-case-sep19-step6.dentocase. Established evidence: collision-aware J1-J5 PreEntry endpoints existed, but Home-to-PreEntry P1 connectivity failed; P2/P3 did not run. J6 and PreEntry-IK theories were not to be reopened without contrary evidence.

Narrow diagnostic change

Experiment A required a checked-in copy of canonical dentobot.urdf with exactly the complete pneumatic_spindle-Copy_collision block removed. One default-off simulation option selects it before robot_description is loaded, shared by MoveIt, guard, robot-state publication and SlicerROS2. No runtime XML surgery, allowed-collision exemption, geometry tuning or generic harness was authorized.

Case loading and profile confusion

The operator encountered Step-6 environment/base_setup_fingerprint post-hydration mismatch, grayed ROS controls, and robot-profile incompatibility. The conversation challenged an exclusion workaround after adding the diagnostic URDF altered resource fingerprinting. A subsequent launch handoff issue meant the first purported diagnostic trial still ran canonical housing-on. These are integration/identity issues, not evidence that the housing-off hypothesis failed. The user created a fresh Sep-22 case; old cases remained historical, not silent fallbacks.

Visible experiment outcomes

Housing-off P1 subsequently passed with 150 waypoints and is labelled M1-DIAG / HOUSING-OFF PASS. It does not close canonical housing-on M1 or authorize housing-off P2/P3. A later close-Home / 45.96-mm opening trial had no tolerance-valid PreEntry IK solution. Another Home at the same gap passed P1/P2 but remained blocked at Stage 3. Base placement/orientation was then changed and an operator screenshot showed a housing-on full chain: 66 / 10 / 22 waypoints, CompletedWithWarnings.

What that history establishes

The original difficulty was not a single immutable algorithm limitation. Setup influences reachable branches and collision geometry. But changing several inputs across these trials prevents causal attribution to one variable. The operator conclusion that physical space mattered is supported as a working interpretation, not a proof that the spindle is globally too large or that changing the base alone always solves it.

4. Chronology II - multi-target memory and activation

The requested workflow

After FDI31 simulation, the operator created an FDI21 trajectory and guide in the same case. Step 5B initially blocked creation with “Choose one current-target trajectory or an explicit pair.” That was repaired sufficiently for the operator to proceed; Step 5C then repeatedly failed “Single current-frame target dock.”

Why this mattered beyond a button

A guide selector alone was insufficient. Selecting or rebuilding one target made the other stale because working Step 4A/4B/4C state and lineage were not restored or scoped correctly to each PreparedBranch. The repeated FDI21/FDI31 alternation exposed ownership, exact trajectory-set, frame provenance and docking-parameter dependencies. Staleness was useful when geometry genuinely changed; it was wrong when merely inspecting an independent prepared branch invalidated another.

Implementation direction retained

The branch must own its exact template, trajectory references, docking assembly, normalized docking parameters and relevant provenance. Activation restores branch working state together, transactionally, rather than copying geometry into a second store or disabling stale checks. Shared case anatomy, jaw opening and robot environment remain shared dependencies. A real change to those dependencies can legitimately stale multiple branches.

Observed milestones and remaining caution

The conversation records repeated failures, then FDI31 Step 5C pass, further FDI21/coexistence corrections, user verification of recent implementation, and a headless Step 4A-through-5C alternating sequence with Step 6A activation and save/reopen. The package FDI21-31-headless-verified-sep22-step6a.dentocase retained both slots Current on fresh reopen. This does not restore live ROS, Home or workspace validity.

Stale diagnostics were a separate issue

The user then ran FDI21 and saw no PreEntry IK solution while the diagnostics window showed a complete chain. The window explicitly said State: Stale; stale reason: Target tooth changed. It was legacy FDI31 evidence, not a fresh FDI21 success. A later claimed FDI21 manual pass screenshot had the same warning and no current Approach plan. These reports must stay visible but cannot be counted as current-target acceptance.

Other requests preserved in context

The user also reported Step 4A scroll-wheel and assisted-trajectory regressions, requested a 10-mm rather than 15-mm Step 4C center-to-dock default, and asked for logbook/screenshot preservation and commits. This report does not independently certify each historical edit or commit; these are related requests, not explanations for the current native Stage-3 contact.

5. Chronology III - controls, batch implementation and evidence

Planner observability first

The operator requested visible planner identity, a chooser, approximate-IK and Cartesian controls, timing/attempt tips, and shared parameters for Approach (6.5) and Drill (6.6). RRT* was added after the initial chooser omitted it. Tooltips explain locked controls and parameter usage. Approximate IK remains disabled; Cartesian Stage 2/3 remains enabled and locked until an independently checked replacement exists.

Retained failure text

The user closed a detailed result dialog without saving it and asked to recover the text in Motion Diagnostics. The retained error-dialog text panel was implemented and the operator confirmed it, along with tooltip explanations. Screenshots show Stage 1/2 pass and Stage 3 guard failure even though geometric Cartesian completion is 100%.

One click, not three manual runs

The user rejected three separate planner operations for every tooth. The approved batch plan requires an identical frozen identity, fresh transient guard/planning context for each trial, cancellation between trials, full diagnostics and exact paths, branch-isolated storage, and non-authorizing replay on reopen. The single-planner control remains for later amendment, with a new live plan required.

Source checkpoint

The Sep-23 logbook records implementation in DENTOSTep6State, DENTORobotWorkflowFacade, DENTORobotSimulationPanel, parameter_state, runtime, widget_robot and widget_robot_shell, plus focused tests. Initial recorded pure suite: 27 passed. Following identity repair: 28 passed. Later timing/source checks were targeted. These are historical recorded results; this report did not rerun the production suite.

Case correction

An early request said FDI21 and FDI32, and a filename also contained "32." Inspection showed PreparedBranches for FDI21 and FDI31; the operator explicitly corrected the target pair. The authoritative input became FDI21-31-headless-verified-sep22-step6a.dentocase. Filenames are not target authority; branch lineage determines the actual target.

6. Runtime attempt ledger - why completion is still open

Run	Outcome / interpretation
FDI21 attempt 1	Stopped before planning: local MRML robot not loaded; ROS2 module startup warning. Automation setup failure, not planner failure.
r2	Robot and scene loaded; saved Task Home rejected as belonging to different robot resources after OMPL profile change. No trial.
r3	Explicitly approved production Home recapture saved a new current-profile case. Comparison then failed identity check; all rows NotRun. Screenshot initially showed main window, not report.
r3 source diagnosis	Scene audit generated_at_utc changed during identical re-acknowledgement. Batch had frozen event identity as content identity. Added stable content fingerprint and retained per-trial audit identity.
r4	RRTConnect reached Stage-3 tooth/spindle guard failure. RRT and RRT* failed prerequisites because identical scene re-acknowledgement invalidated Home. Not a valid three-planner comparison.
r4 source correction	Preserve prior audit identity only if all fields except timestamp/fingerprint match after fresh acknowledgement. Changed payload/policy/status still invalidates.
r5	Hit 1200-second cap; no completed result or saved comparison case. Server logs show planner configurations, but no complete per-trial outcome evidence.
r6	Timing instrumentation and per-trial checkpoint retained completed RRTConnect Stage-3 collision. Guard cost dominates. RRT interrupted; RRT* no completed trial; no full comparison case.
Offline capture 1 / 2 / 3	First unsupported camera API; second omitted spindle in framing; third showed separated tooth/spindle and was rejected. Source later found upper-tooth jaw-transform defect.

Each run above had its own authorization and recorded stop. Run numbers do not reset a causal retry ceiling. Later FDI31 and completed-record offline reopen checks did not occur merely because a partial output case existed.

7. Current blocker - exact FDI21 interpretation

Latest meaningful trial

FDI21 r6 used the canonical housing-on URDF and the separately saved current-profile Home case. The retained representative RRTConnect candidate is index 6, workspace clearance sample 198. Home-to-PreEntry and PreEntry-to-Entry succeeded. Full-task Stage 3 was rejected at composed waypoint 459, drilling sequence 460, interpolation sample 1/1.

Named pair

dentobot_target_tooth_2.25.156727832881888777359082987559036393993 versus pneumatic_spindle-Copy. This is target anatomy versus spindle housing, not merely permitted burr-to-target contact. The guard rejection must remain a blocker under current policy.

Why “100%” is not success

Cartesian completion describes how much of the requested path was generated kinematically. Independent full-chain guard evaluation can reject a point on that generated path. Thus a Stage-3 fraction of 100% with Stage-3 Failed is coherent: the requested endpoint was represented, but the path was not accepted. “Approach ready” is not Drill authorization.

What is and is not known

Known: native guard emitted a non-approved pair at a recorded state; one candidate path and its failure report were retained. Unknown: contact depth/nearest point in r6, exact physical explanation in a corrected visual reconstruction, whether another admissible tool frame/base arrangement avoids the contact, and whether all meaningful arrangements are infeasible. One failed candidate is not a global impossibility proof.

Independent envelope warning

Requested drilling depth 7.571 mm plus provisional guide/shell traversal allowance 0.500 mm gives 8.071 mm combined insertion. The provisional envelope is 6.500 mm: excess 1.571 mm. Effective modeled tool protrusion is 7.000 mm with a 0.500-mm reserved visible/protrusion margin. This warning is distinct from native collision. It should not be erased by shortening the approved trajectory or increasing an uncalibrated number until green.

Safety interpretation

These are research-simulation quantities, not measurements of the actual handpiece, guide or patient. The report makes no clinical fit or safe-drilling claim. A new geometry decision needs the operator’s dimensional and physical interpretation, with the existing collision and phase checks preserved.

8. Why the collision screenshots were invalid

Operator correction

Tarun explicitly rejected the pictures: the spindle and something resembling the target tooth were visible, but the colliding object was not represented in contact. A picture with the right-looking components is not diagnostic evidence unless it depicts the same object identities, transforms and robot state as the native query.

Concrete source finding

Testing/run_dentobot_step65_collision_recapture.py obtains the target surface, then unconditionally calls _step6CaseJawPolydataWorld. Production logic_robot_scene_sync.py starts with the original collision surface and applies that transform only when the segment belongs to the moving lower jaw. FDI21 is in the fixed upper set; its saved audit records classification fixed and jaw_transform_application_count = 0. The recapture therefore incorrectly moved the upper tooth.

Numerical corroboration

Saved correct target world-RAS bounds, mm: X [-99.980, -91.867], Y [-44.774, -33.631], Z [53.930, 77.450]. Invalid recapture: X [-99.511, -91.399], Y [-65.894, -54.356], Z [18.623, 41.046]. Displayed spindle: X [-106.345, -85.794], Y [-40.817, 22.878], Z [38.431, 75.941]. The invalid target has about 13.54 mm Y separation from the spindle. Correct saved target bounds overlap the spindle box in all axes.

Evidence boundary

Bounds overlap is only a broad spatial consistency check, not triangle contact, penetration or native collision proof. The saved case audit also precedes the r6 acknowledgement, so matching its geometry does not alone certify the r6 native scene. The repair must reconcile the actual collision object/payload, frame and exact evaluated state, not just produce a visually persuasive overlap.

Immediate source step, not yet done

Use existing fixed-upper/moving-lower classification or the exact audited collision payload in recapture. Do not add a second transformation policy. Check fixed upper remains untransformed and lower receives the opening exactly once. Preserve source anatomy. Reconstruct only display-only evidence, record object IDs and state identity, and fail closed if the required pair/state cannot be established.

Current protective change

A source-only bounds-overlap check was added to the capture paths to prevent separated models being labelled collision views. It is not runtime-verified and does not solve scene attribution. The known unconditional-transform defect remains present at report preparation. No fourth recapture was run for this report.

9. Why 5 seconds became a 20-minute run

Different levels of time budget

The 5.0-second parameter applies to an individual native joint-planning request. One Approach trial can enumerate multiple IK seeds, attempt direct and clearance routes, perform Cartesian preflights, synchronize/prepare guards, and validate full waypoint sequences. Three configured planners repeat this higher-level work. The runner's 1200-second cap is an outer process limit, not the planner setting.

Measured r6 timing

Case opened at about 60.4 s; Step 5C verified at 66.5 s; branch activated at 70.5 s; ROS connected at 107.0 s; Task Home validated at 109.5 s. The completed RRTConnect checkpoint appeared at 1050.4 s. Its trial took about 874 s. Eight completed guard_waypoints calls totalled 813.39 s, largest about 141.8 s. Fourteen joint-plan calls totalled about 4.6 s; fourteen Cartesian calls about 21.97 s.

Interpretation

Guard work was approximately 93% of that trial's measured duration. These nested instrumentation durations are not independent buckets that should all be added. The evidence points to waypoint validation as the first performance investigation, not simply a slow RRTConnect solver. It does not yet prove which lower-level transport, synchronization, collision query or repeated computation dominates each guard call.

20 minutes versus 30 minutes

Raising the cap to 1800 seconds allows 600 more seconds of work before termination. It could permit more trials to finish, but does not remove collision, fix identity, make bad pictures valid or guarantee completion. It may only spend longer repeating rejection. No automatic cap increase is approved by this report.

Options already discussed

Timing option A: add flushed UTC and monotonic phase/call records and checkpoints; implemented and run as r6. Timing option B: impose a total per-planner trial budget; operator rejected it. Timing option C: optimize only the measured repeated work after evidence, preserving exact validation semantics; not yet implemented. These option names are distinct from historical Task Home Experiment B and the later rejected manual-development Path B.

Safe performance direction

First attribute cost at the relevant guard boundary; then remove redundant setup/transport or repeated invariant computation only if equivalent inputs and fresh authoritative checks are preserved. Do not cache collision verdicts across changing state, skip waypoints, loosen tolerances, reduce search silently or claim a fair comparison after changing one planner's budget.

10. Base, Task Home and mouth opening - causal roles

Robot base / mount

The base transform changes the target path expressed in the robot frame and can change joint-range feasibility, conditioning, available branches, upstream-link clearance and reachable tool orientation. A small position or orientation adjustment can matter greatly in a constrained five-joint system. A mount proposal must still be anatomically/mechanically admissible; an arbitrary translation is not a clinical mounting solution.

Task Home

Home changes the initial configuration, approach connectivity and IK seed/branch selection. It is a strong candidate variable when valid PreEntry endpoints exist but P1 cannot connect. If P1/P2 already pass and Stage 3 collides, Home is not automatically the primary lever. It might indirectly select another branch/tool frame, but does not change contact at an identical full world-space housing pose.

Mouth opening

Increasing the gap moves lower-jaw anatomy and can improve access around it. Upper FDI21 is fixed in the case's head/world representation. A larger opening therefore does not itself move FDI21 away from the housing; any improvement must come through changed surrounding access or another setup/path change. The measured 45.96-mm gap is an input to study, not a general prescription.

Axis versus complete orientation

Position-axis IK constrains TCP position and drill axis using J1-J5. Different admissible arm solutions may place non-axisymmetric housing differently around that axis. A base or seed change can alter which solution is found. This is a reason to inspect exact FK and housing pose, not authorization for speculative roll sweeps, J6 planning, new DOFs or relaxed residuals.

What the operator's successes contribute

The altered FDI31 full-chain pass is a valuable feasibility witness for its recorded visible setup. Its role is to motivate a causal investigation and, if the exact state is later available, a controlled reference. It cannot become a universal base/Home rule or substitute for current FDI21 evidence.

Required form of future advice

Every proposed adjustment should state: failed stage, exact observed constraint, why the variable can affect it, what stays fixed, allowable adjustment bounds, what result would refute the hypothesis, and which evidence the operator must review. "Try moving the robot" is not an adequate Studio diagnostic.

11. Differential diagnosis - do not merge the failures

Observed symptom	Interpretation and discriminating evidence
No tolerance-valid PreEntry IK; collisions empty	No solution found by the bounded solver/seeds at this setup; not proof of collision or global unreachability. Check residuals, joint margins, frames and branch search evidence.
Valid PreEntry endpoint; P1 fails	Endpoint validity is not connectivity. Examine start/goal, direct-segment first collision and bounded native routes. Home/base may matter for different reasons.
P1/P2 pass; P3 native contact	Task-line/branch/housing feasibility is the immediate question. Changing free-space planner cannot by itself remove the same downstream contact.
100% Cartesian fraction; guard fails	Generated path reached its requested extent; independent acceptance rejected it. Do not promote partial/invalid path to Drill preview.
Old complete diagnostics after target switch	Historical stale evidence, not a fresh pass. Match branch, trajectory, task and base fingerprints, not visible tooth labels alone.
Case/profile/Home rejection	Resource or lineage prerequisite failure, not an algorithm result. Do not disable compatibility gates or call NotRun a planner failure.
Separated objects in contact screenshots	Evidence reconstruction/frame mismatch until resolved; cannot overturn native collision or justify geometry changes.
Timeout	Incomplete computation/evidence, not infeasibility. Inspect the last flushed phase and completed checkpoints.

12. Recommended path A - finite, evidence-led gates

Gate A1: repair attribution at the smallest scope

Correct the identified upper/lower transform defect and add one focused check. Reuse existing scene classification or audited outgoing geometry. Verify source/target IDs, units and frame chain. Do not rerun three planners to learn whether a fixed upper tooth was wrongly transformed.

Gate A2: reconcile one retained rejected state

Under a separately approved bounded runtime scope, reconstruct or evaluate the retained first-invalid state against the matching scene. Capture context, pair close-ups and the same state from inferior, apical and oblique views. Preserve joints and audit identity. Outcomes: agreement and contact leads to feasibility reasoning; mismatch leads to the precise scene/state defect; missing data leads to a named instrumentation request, not another blind full run.

Gate A3: establish insertion feasibility

Inspect the actual approved Entry-to-Target line, housing/burr/guide dimensions, fixed tool frame and relevant joint margins. Separate target contact, upstream collision, limit/reach failure and provisional envelope warning. Determine whether the current branch fails and whether another admissible arrangement is plausibly available. Do not turn a bounded negative search into a universal impossibility claim.

Gate A4: constrained adjustment only when justified

If evidence identifies a base/Home mechanism, define one controlled comparison with the operator: fixed case/trajectory/policy, one declared variable, admissible range and a discriminating result. Preserve failed and successful configurations. Seek a useful feasible region and margins, not an unexplained single green pose. Manual expertise supplies constraints and interprets geometry; it is not a bypass.

Gate A5: measured efficiency and batch completion

Once attribution and the relevant feasibility question are settled, implement a bounded optimization of measured repeated guard work if needed. Complete the same-input planner comparison, save a new case, reopen offline and inspect reports/ghost paths while guarded preview stays gated. Only then assess relative behavior; do not label a planner better because a common Stage-3 constraint blocked another trial.

Stop rule that prevents spiralling

Each action must discriminate named hypotheses and produce a decision. An unchanged unexplained failure is not permission to increase budget, add candidates, move geometry or repeat the entire workflow. Stop at the first decision-critical unknown and return a usable evidence packet with a specific operator question. The three-failure ceiling and manual verdict gate remain.

13. Rejected paths and valid alternatives

Rejected development Path B

The recent alternative was to manually adjust the setup until it passed, save a “golden case,” and use it to unblock the rest of the workflow. Tarun rejected this as manipulating values to obtain a desired result. It is not recommended as the main strategy because it confounds variables and provides weak generalization to other teeth or mounting constraints.

What manual work remains useful

A deliberately documented manual observation can be excellent evidence. A saved successful state can establish an existence witness; a before/after pair can test a named mechanism. The boundary is causal discipline: retain exact inputs, do not discard failures, preserve geometry/guard policy, and never assume a manually convenient mounting pose is physically admissible.

Alternative if A exposes an actual design constraint

If corrected evidence shows a geometric/tool constraint for the reviewed setup, the appropriate next step is an operator-led design/input review: measured tool protrusion, housing envelope, guide dimensions, mount feasibility or the approved trajectory. That is an explicit changed experiment, not an invisible numerical workaround. The software should report the constraint honestly and preserve the prior result.

Alternative if evidence remains insufficient

If the exact native state or scene cannot be reconstructed, keep the run Inconclusive and instrument the missing boundary. Do not infer contact depth from pixels or manufacture a “collision-free” result by suppressing a body. A truthful inability to decide is preferable to a persuasive but misregistered image.

Does A guarantee success?

No. It should guarantee a bounded improvement in knowledge: identify and fix a software defect, demonstrate a constraint for evaluated inputs, justify a specific setup change, or identify missing evidence. It avoids endless trial only if those stop conditions are enforced. A planner cannot promise a feasible path through an impossible geometry.

14. Implementation map and remaining source boundaries

Existing comparison and persistence

DENTOSStep6State.py owns record/integrity helpers; DENTORobotWorkflowFacade.py runs the ordered comparison through existing Approach planning and retains reports/paths; DENTORobotSimulationPanel.py and widget modules expose comparison, shared parameters and saved diagnostic views. Parameter/runtime state carries branch-keyed records. These source changes are in the dirty working tree and cannot be reproduced from HEAD alone.

Existing automation

Testing/run_dentobot_step65_three_planner_gui.py drives the production flow, retains setup and per-trial checkpoints, writes timeline.jsonl, captures diagnostics and has a fresh-process reopen mode. It must not select a replacement target by filename, silently repair Home, move the base, run preview or bypass eligibility. Explicitly approved Home recapture was a separate, narrowly gated step.

Immediate edit candidate

Testing/run_dentobot_step65_collision_recapture.py contains the unconditional jaw transform. The smallest meaningful repair is reuse of the production fixed/moving semantics, with a focused upper/lower check and evidence identity validation. The production planner should not be changed to compensate for a diagnostic renderer bug.

Exact guard-state preservation

The native guard and DENTOROS2Bridge status already expose evaluated_positions, sample index/fraction, first/second body and world-object evidence. The facade currently retains a waypoint-level requested vector. At the reported sample 1/1 this is the endpoint, but a general interpolated rejection can differ. Reuse the existing native fields if exact-state persistence is needed; do not invent a second state evaluator. Contact/nearest fields are null in r6 and must remain unknown.

Later work, not bundled into this repair

Cartesian-off primary sequential exact-pose IK, full-cycle Drill/Return Home acceptance, broad Studio scheduling, automated mount optimization and model/resource migration are separate scoped work. No new planner framework, XML-transform framework, database, persistent evidence service or generic harness is needed to fix the known capture error.

15. Studio foundation - what should survive this work

One authoritative case, distinct branches

Keep .dentocase as the package authority and MRB as geometry authority. PreparedBranches own their target-specific guide/trajectory/provenance; shared environment changes explicitly invalidate dependent evidence. Switching selection alone must not overwrite another branch's stored result. Stale evidence stays inspectable and labelled.

Immutable attempt evidence

Each comparison needs exact branch/task/base/Home/scene/profile identities, settings, requested/configured planner IDs, stage outcomes, exact failure text, complete or last-valid full-precision paths and stage boundaries. Record timing and screenshots at completion of each trial so a later timeout cannot erase earlier findings.

Offline review is not runtime restoration

Saved paths can produce transient ghost replay at their saved coordinates when integrity and robot/base identity match. They must not move the authoritative robot, publish collision objects, reconnect ROS, restore Home validity or become a guarded plan. A selected planner still needs a fresh current live plan through all gates.

Dentist-facing result categories

A useful result explains whether the evaluated task is feasible in the reviewed setup, blocked by a demonstrated constraint, not demonstrated under bounded search, or inconclusive due to missing/stale evidence. Advice should separate mounting, Home connectivity, local tool clearance, guide/tool capacity and data-quality problems. The operator should not need to infer these from a generic red error box.

Scaling boundary

The current registry vision is 32 teeth x 3 trajectory slots; "up to 100" is the broader product aspiration. A batch may share environment setup only while identities remain unchanged. It must not silently move the base per target and call the study one fixed-setup result. The scheduler follows stable single-task semantics and full-cycle acceptance, rather than multiplying current ambiguity by 96.

16. Decision checklist before authorizing further runtime

1. Agree the immediate question

Is the current native FDI21 rejected contact represented correctly in evidence? Not: can another planner eventually turn the task green?

2. Freeze the scope

Source repair and focused check first; then one exact-state diagnostic if approved. No three-planner rerun, automatic base/Home movement, tolerance change or new geometry.

3. Define acceptable evidence

Same named pair, matching geometry/frame identity, exact rejected state, three usable angles and context, native contact fields where available, unknowns explicitly labelled.

4. Define the branching decision

If evidence matches: discuss the physical/kinematic mechanism. If not: repair the exact mismatch. If incomplete: collect only the missing discriminator.

5. Preserve the operator's role

Tarun interprets the geometry and admissible setup constraints. An approval request must include evidence and a concrete choice, not merely ask to continue.

6. Do not conflate milestones

Source checks do not close runtime acceptance; a housing-off pass does not close housing-on M1; an FDI31 witness does not close FDI21; saved replay does not authorize motion.

7. Resume batch only on a reasoned basis

Use the measured cost evidence and accepted setup to choose the next bounded comparison. Preserve all failures and NotRun outcomes; save/reopen the completed record before wider Studio work.

No implementation or runtime action was performed while producing this report. The next action remains a decision for the operator.

17. Exact artifacts and reproducibility identifiers

Original Sep-19 FDI31

/home/light-tarun/dentobot/data/Slicer_Saved/SampleStudy1/FDI31/dentobot-case-sep19-step6.dentocase

SHA-256: dfe492076674d9fa34c8eb9e010ff6a69e022438b06a081ea3fc4c6a1fc981ab

Sep-22 FDI31

/home/light-tarun/dentobot/data/Slicer_Saved/SampleStudy1/FDI31/dentobot-case-sep22-step6.dentocase

SHA-256: eb48a805c81578bafcc8ca72663b6f8a98b1f9721c7f9e680d44b458558e7adb

Two-target headless-verified input

/home/light-tarun/dentobot/data/Slicer_Saved/SampleStudy1/FDI21-31-headless-verified-sep22-step6a.dentocase

SHA-256: c16e0406b589de1cb2144628b38a26100157458d48b5edb9c56ad4f8792d1c1d

Current-profile Home source

/home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r3/FDI21-current-profile-home.dentocase

SHA-256: d6da41b87c63549bdcffef06ce1df779f3ed396fe304ed826010147dfed0510c

Latest completed-trial checkpoint

/home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/checkpoint.json

SHA-256: 49115ceaf8d3cfcf8e72444a81d6ca74e694d41587dfe97e9805be3fdc7883b1

Timing events

/home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/timeline.jsonl

SHA-256: 9502a7eedc5f2937351ac7ecf7447615d5dc6b80b5e79edb60869ff20dc75ccf

Invalid capture manifest

/home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6-collision-recapture-r3/manifest.json

SHA-256: 9ea57c679025ce82188aeb97a6dcac6e9689978db3aa535514d476e4bad1d95a

The hashes above are computed during report preparation. A checkpoint is not a completed case. Source case and run outputs are preserved separately.

18. Retained rejected state and identity

The following state was retained by the offline capture manifest from the r6 checkpoint. The manifest’s tooth geometry is incorrect; the numeric robot vector and native message are retained evidence, not validation of its images.

Joint	Recorded SI value
link-1_Revolute-1	0.042047916865283974
link-2_Slider-2	0.00036656668634758036
link-3_Revolute-3	3.4388198100123732
link-4_Slider-4	0.013680814721990668
link-5_Revolute-5	0.4085713105355186

Revolute joints use radians; slider joints use metres. J6 is external spindle action, not a planned positioning DOF.

Native message

Full-chain guard rejected waypoint 459 (drilling): Task guard rejected drilling sequence 460: MoveIt detected a non-approved collision at interpolated sample 1/1.
(dentobot_target_tooth_2.25.156727832881888777359082987559036393993 ↔
pneumatic_spindle-Copy)

Checkpoint context

Selected candidate: 6; workspace clearance sample: 198. First-invalid TCP world RAS approximately [-95.45838257548839, -37.504953467682284, 60.796333221604584] mm. Nearest points and native contact depth are unavailable in the retained r6 record.

Base fingerprint

230ce2cda5723fbc41fadfd9c1f0f887710b29a444cbf73bee4daddaa394e47c

Source boundaries

The source case audit has the upper tooth correctly fixed. The r6 live acknowledgement has a different audit event identity. Correct source geometry is necessary but not sufficient to assert exact live-scene equivalence. That comparison belongs to the next bounded evidence gate.

19. Source index and evidence hierarchy

S1 Canonical execution contract

/home/light-tarun/dentobot/docs/diagnostics/FDI31_GUI_PLANNER_P0_PLAN_2026-09-21.md

S2 Current run history and source findings

/home/light-tarun/dentobot/docs/logbook/2026-09-23.md

S3 Prior multi-target and manual results

/home/light-tarun/dentobot/docs/logbook/2026-09-22.md

S4 Pending state and owner mapping

/home/light-tarun/dentobot/docs/backlog.md

S5 Task contracts

/home/light-tarun/dentobot/docs/TASKS.md

S6 Milestone / Studio design

/home/light-tarun/dentobot/docs/DEVELOPMENT_PLAN.md

S7 Technical decisions

/home/light-tarun/dentobot/docs/DECISIONS.md

S8 Verification gates

/home/light-tarun/dentobot/docs/AGENTIC_VERIFICATION_PROTOCOL.md

S9 Automation SOP

/home/light-tarun/dentobot/docs/diagnostics/STEP6_GUI_AUTOMATION_SOP_2026-09-23.md

S10 Recapture implementation

/home/light-tarun/dentobot/ros2_ws/src/DentoBot/Testing/run_dentobot_step65_collision_recapture.py

S11 Production scene preparation

/home/light-tarun/dentobot/ros2_ws/src/DentoBot/DENTOWorkflow/Resources/Python/dentobot_workflow/logic_robot_scene_sync.py

S12 Native evaluated state

/home/light-tarun/dentobot/ros2_ws/src/DentoBot/dentobot_moveit_config/src/collision_guard.cpp

S13 Comparison automation

/home/light-tarun/dentobot/ros2_ws/src/DentoBot/Testing/run_dentobot_step65_three_planner_gui.py

Sections 1-5 synthesize the supplied chat plus S1-S7. Sections 6-9 use S2 and retained r3-r6 artifacts, with S10/S11 confirming the image-transform defect. Sections 10-16 distinguish engineering interpretation and proposed decisions from implemented behavior. The atlas captions name each image and its evidentiary limitation.

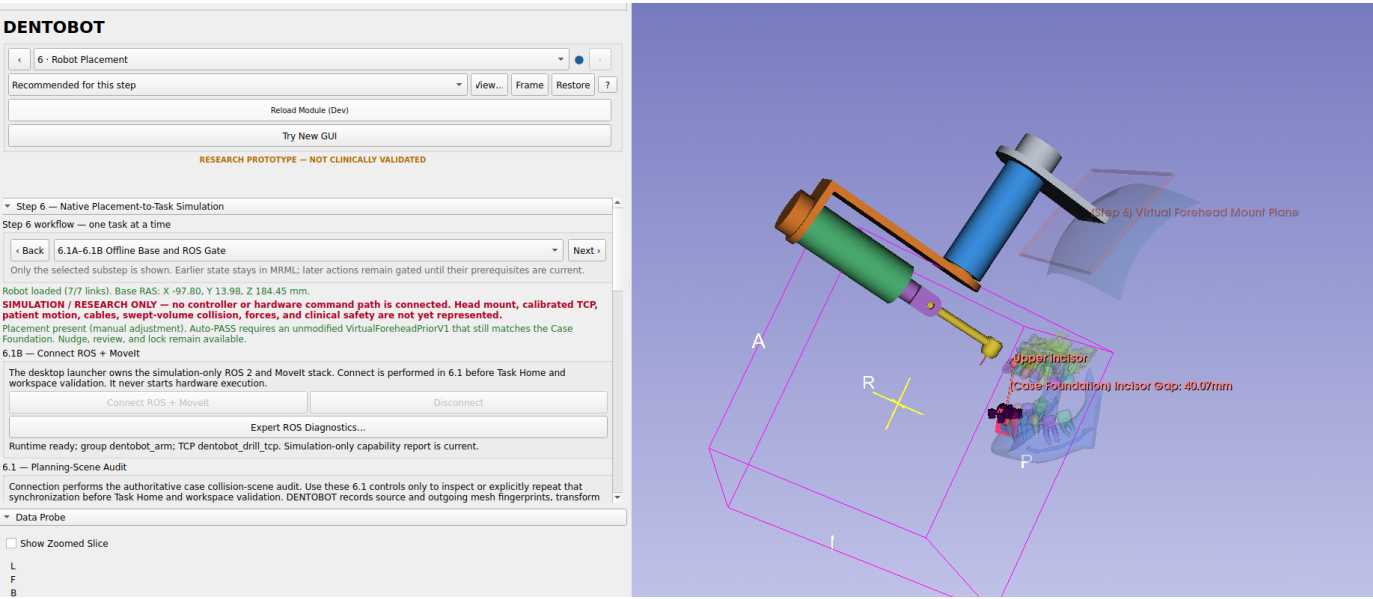
20. Screenshot atlas - interpretation rules

The following images are original user-provided screenshots or retained automated captures. They are embedded without reconstructing UI text or editing the scene. Each image is scaled to fit the page; zoom the PDF to inspect native screenshot detail.

Screenshots are chronological evidence, not instructions. Old successful results do not become current merely because they appear beside a new target. “Stale,” “NotRun,” partial path and guard warnings are retained in the interpretation.

The rejected offline collision images are included deliberately, marked INVALID FOR CONTACT INTERPRETATION. They document the evidence failure and must not be offered as proof of clearance or penetration.

Figure 01 | Early Step 6 ROS connection ambiguity

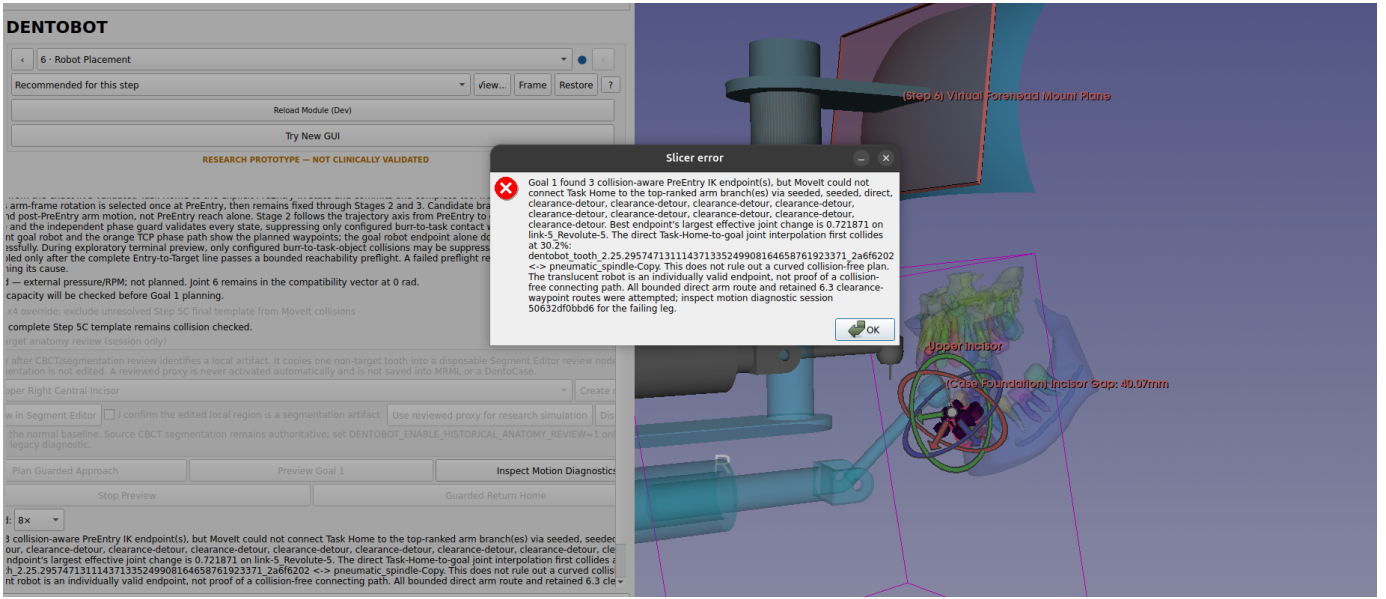


Operator reported both Connect and Disconnect grayed after case loading. The displayed Runtime ready message does not explain the gating state; this is UI/prerequisite evidence, not a planner outcome.

Source: /tmp/codex-clipboard-6f55b869-1db5-4e0d-a162-2b7d67a6620f.png

Image SHA-256: fec3669cd4932a3544e535a10574bea3e0ffc2d37588bbcf134c401466b146f7

Figure 02 | Sep-22 FDI31: endpoints exist, P1 connection fails

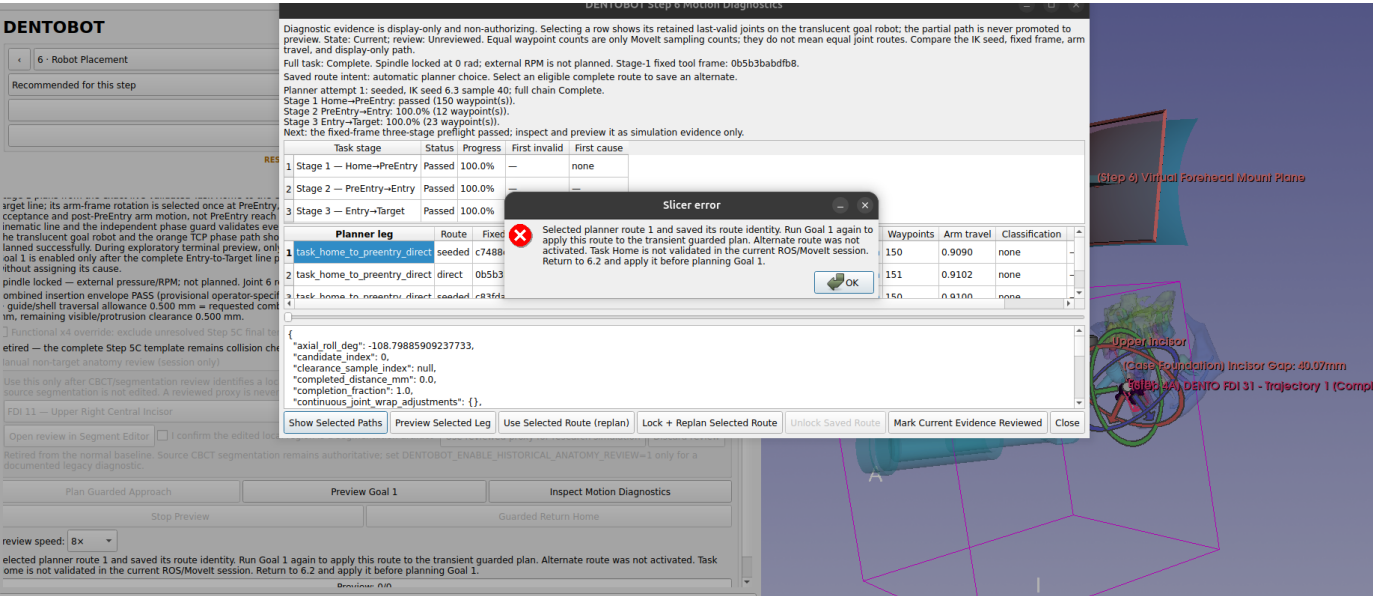


Three collision-aware PreEntry endpoints were found. Direct interpolation reports tooth versus spindle contact. The first Sep-22 attempt still used housing-on due to launch handoff; this is not proof Experiment A failed.

Source: /tmp/codex-clipboard-f0ef4418-c2c2-4b8d-bf54-348a7d58d4bf.png

Image SHA-256: 21e157c4b6e265c1f06449ce54e02cc95c2d7e9cd8f80f5a18077495251cf126

Figure 03 | Diagnostic phase pass and route-activation gate

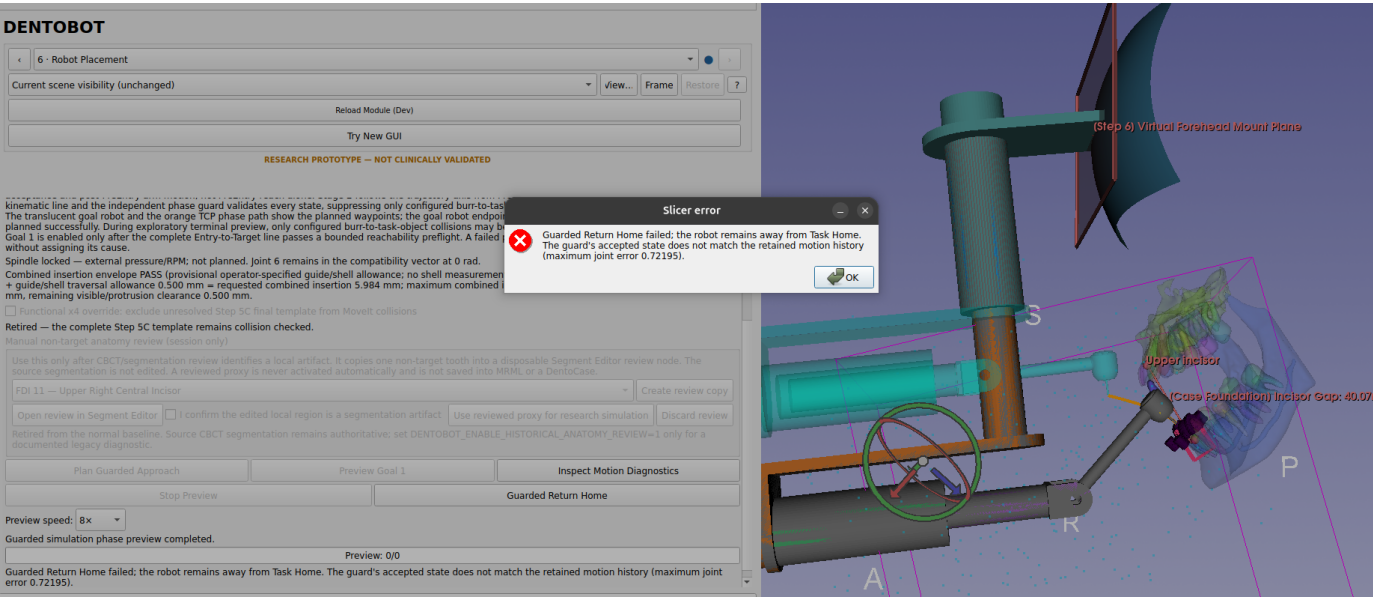


Stages show complete diagnostic output, but selecting a route reports Task Home not validated in the current session. Historical housing-off P1 evidence is M1-DIAG only; route selection does not restore current authorization.

Source: /tmp/codex-clipboard-41aa5537-4213-43a6-8f87-e531cec7edd4.png

Image SHA-256: fadc5eb7f834b8212f7c07a29fc33d332703371a27776a8cf6a5881b75c787bf

Figure 04 | Guarded Return Home mismatch

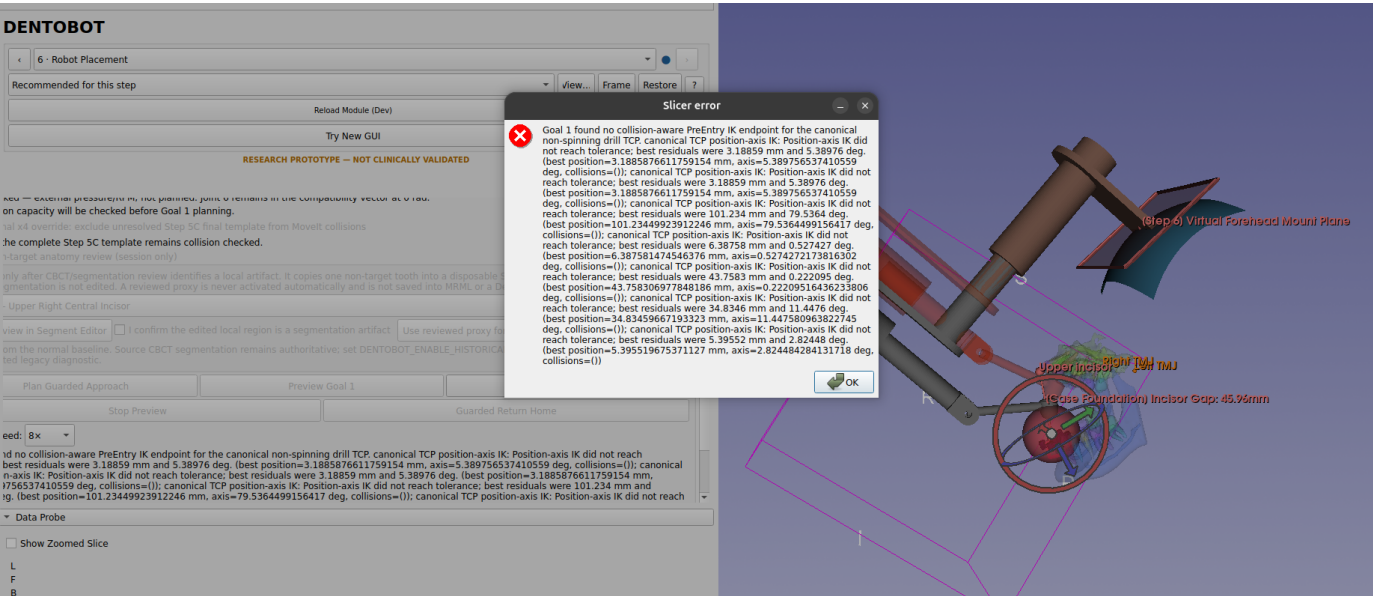


Guard accepted state differs from retained motion history; maximum joint error 0.72195. This is a separate full-cycle state/history issue, not an explanation for all planning failures.

Source: /tmp/codex-clipboard-c8891b3b-878b-4172-b6e9-be8134be261c.png

Image SHA-256: 0436acccaa3198d51fc69fe4c92818786f9629702142a2ce120422f2a8d366fa

Figure 05 | 45.96-mm opening: no tolerance-valid PreEntry IK

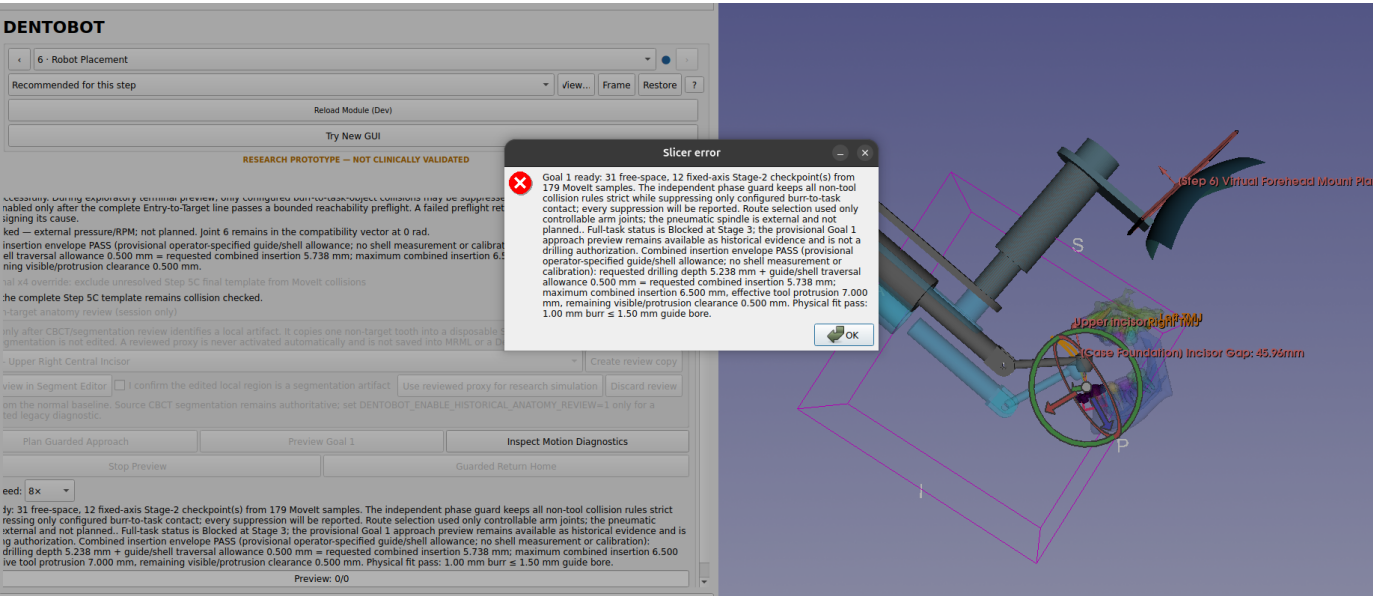


Reported position/axis residuals with empty collision lists. This particular failure concerns bounded IK convergence/reach at that setup, not a demonstrated spindle collision.

Source: /tmp/codex-clipboard-bce2e0f1-ac73-4392-9561-9af7573abad8.png

Image SHA-256: de6ee0346683d20d81e303edbffd87e2d66fa81d79b24152f666d5933488c60d7

Figure 06 | Same gap, changed Home: P1/P2 progress

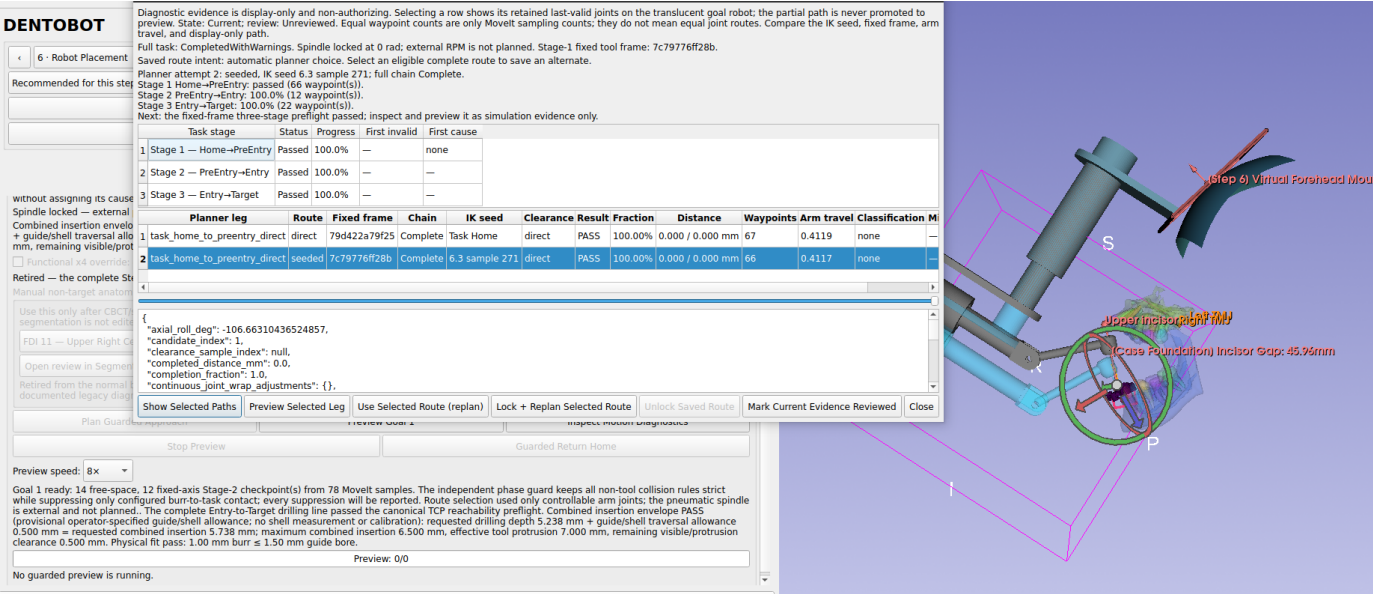


Operator changed Home; approach was available while full task remained blocked at Stage 3. This supports separating approach connectivity from insertion acceptance.

Source: /tmp/codex-clipboard-8c72bf66-55ff-4c8f-afd8-56e7f5b85969.png

Image SHA-256: 0d3c5b738920bdc6f905dfb94167581f6b0ef20cd6b00c8654d6ded3a31c9c7a

Figure 07 | Operator FDI31 housing-on full-chain witness

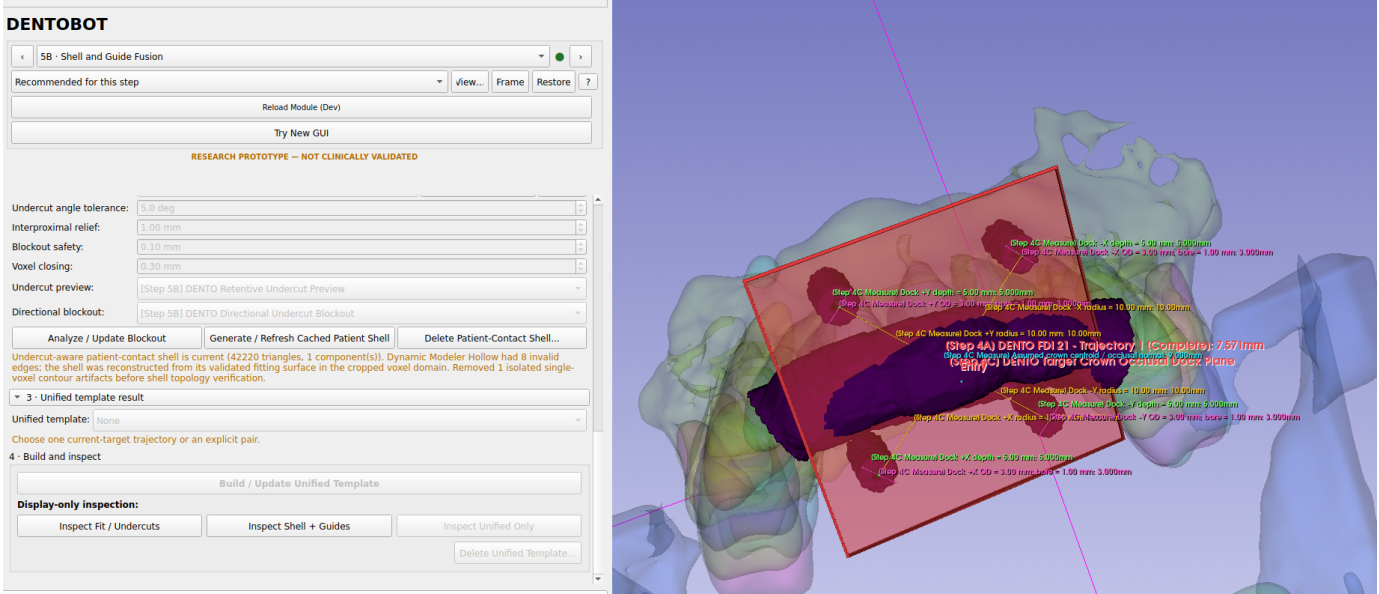


Operator reported base position/orientation adjustment. The display shows 66/10/22 waypoints and CompletedWithWarnings. Meaningful alternate-setup evidence; not an independently reproduced fixed-input experiment.

Source: /tmp/codex-clipboard-5db5f538-f922-418b-b6dc-5c359af02909.png

Image SHA-256: f6e1e1b0a72bd131f62a7d42d4ba6b7551f913e93144f9446401f52b31219359

Figure 08 | Second target blocked at Step 5B

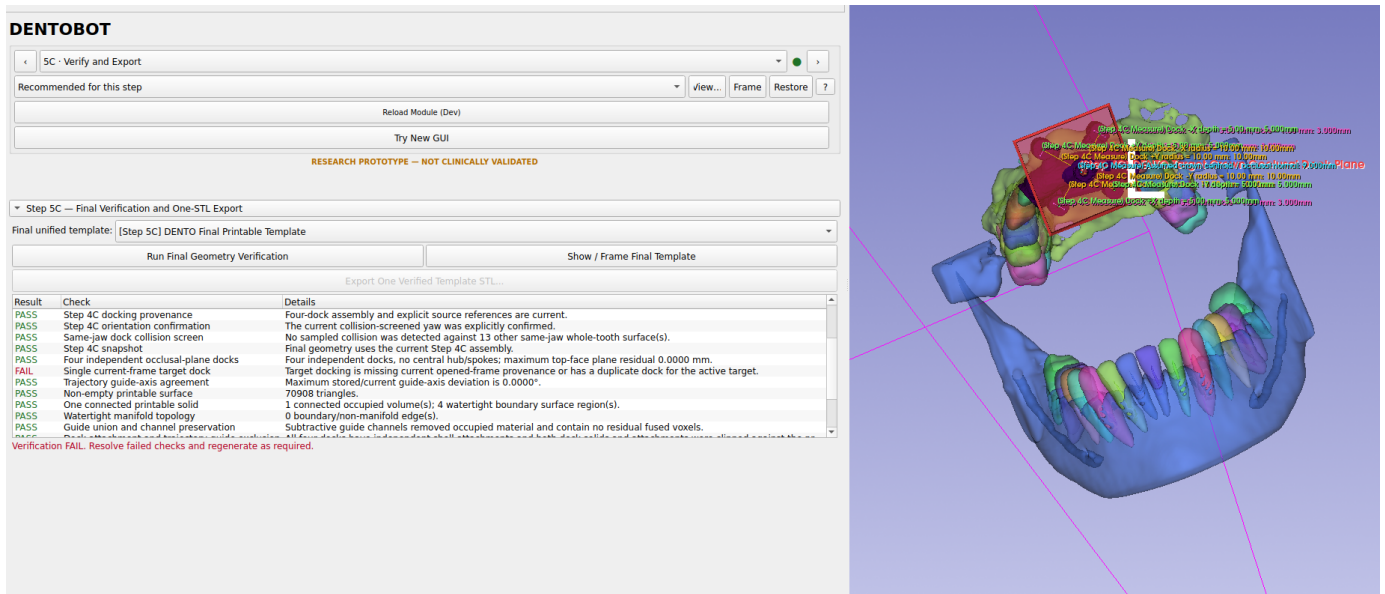


Choose one current-target trajectory or explicit pair prevents unified template creation. This motivated selected-target lineage work.

Source: /tmp/codex-clipboard-ccd43578-a988-4765-a5ec-a8778000768b.png

Image SHA-256: 8220ff2303c6631c855b4f75f5bc25e837606d7e7f2634a3744ba3a15709b208

Figure 09 | Step 5C current-frame dock failure

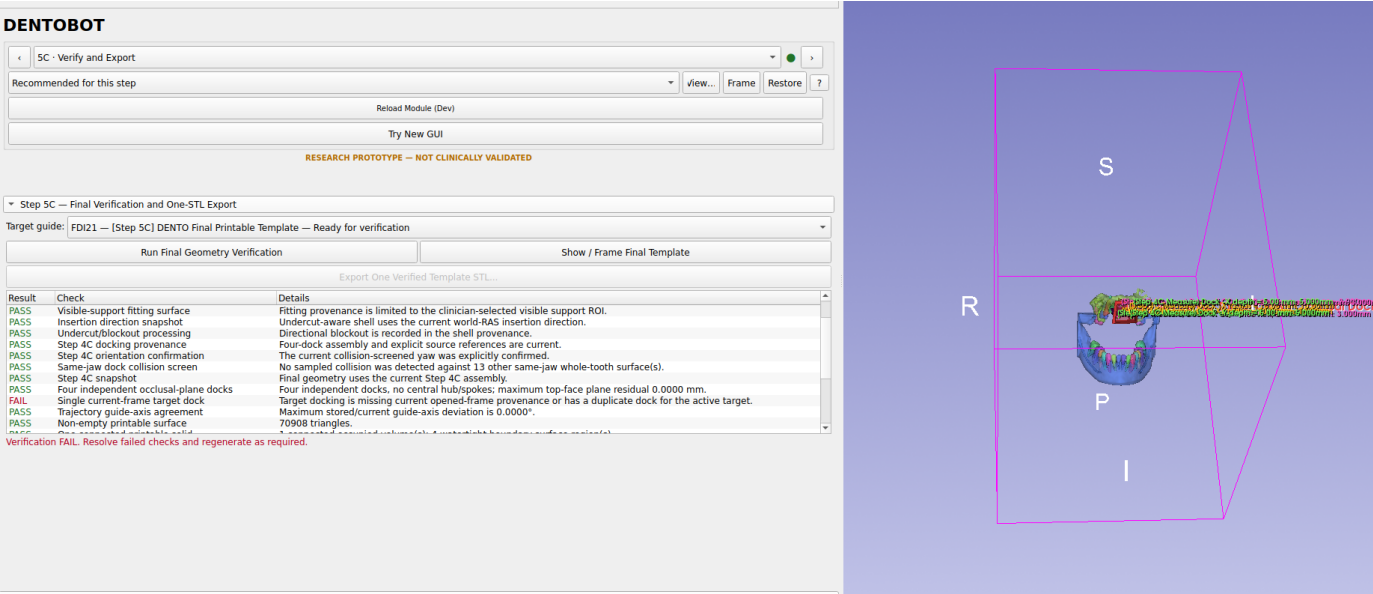


Other geometric checks pass while single current-frame target dock fails. A template's geometric existence is distinct from its current provenance.

Source: /tmp/codex-clipboard-320076a0-a823-4483-bd0e-6cdc4de79d07.png

Image SHA-256: 12d2b747c6baddcf3699e44d6f858d6ffda58013a52921bea522c76d3a78f502

Figure 10 | Selector added, verification still failing

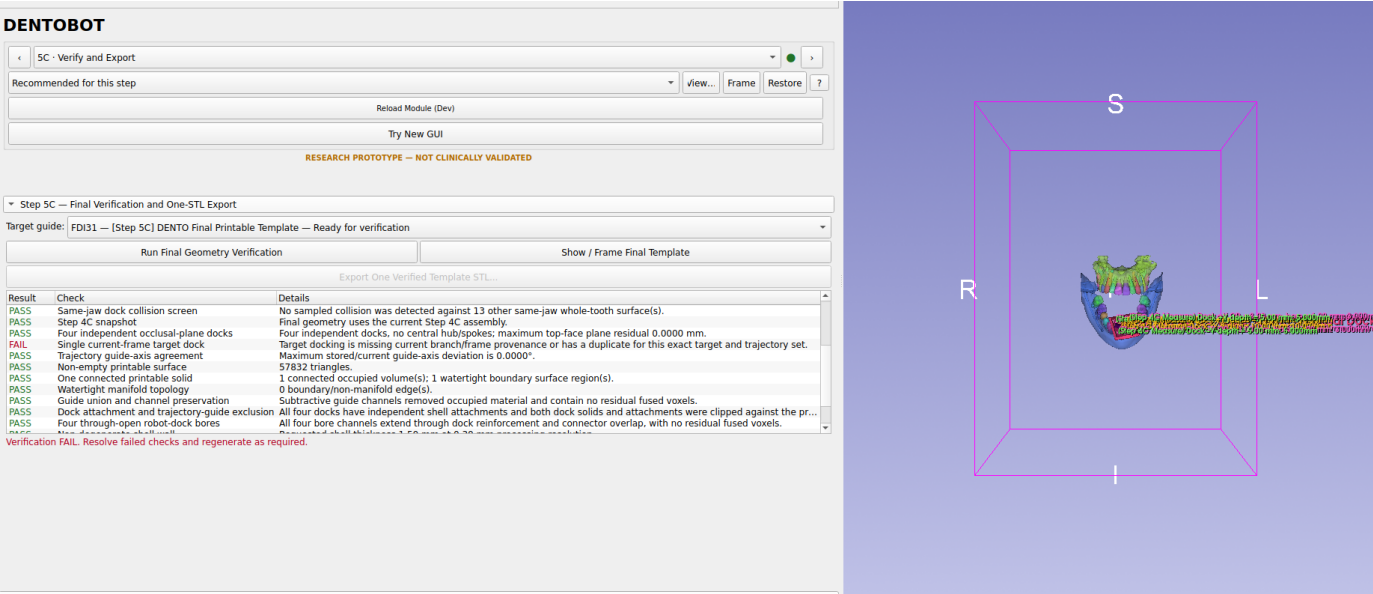


FDI21 Ready for verification did not imply successful verification. The operator correctly challenged a UI change that did not resolve the underlying lineage check.

Source: /tmp/codex-clipboard-72314d4f-19b2-4628-9c00-4bc4eeebd25d.png

Image SHA-256: ee76e9f28342acbadc17a92ab51955d62597f965736b2d82264e0c564e75b0c6

Figure 11 | Rebuilt FDI31 still fails dock provenance

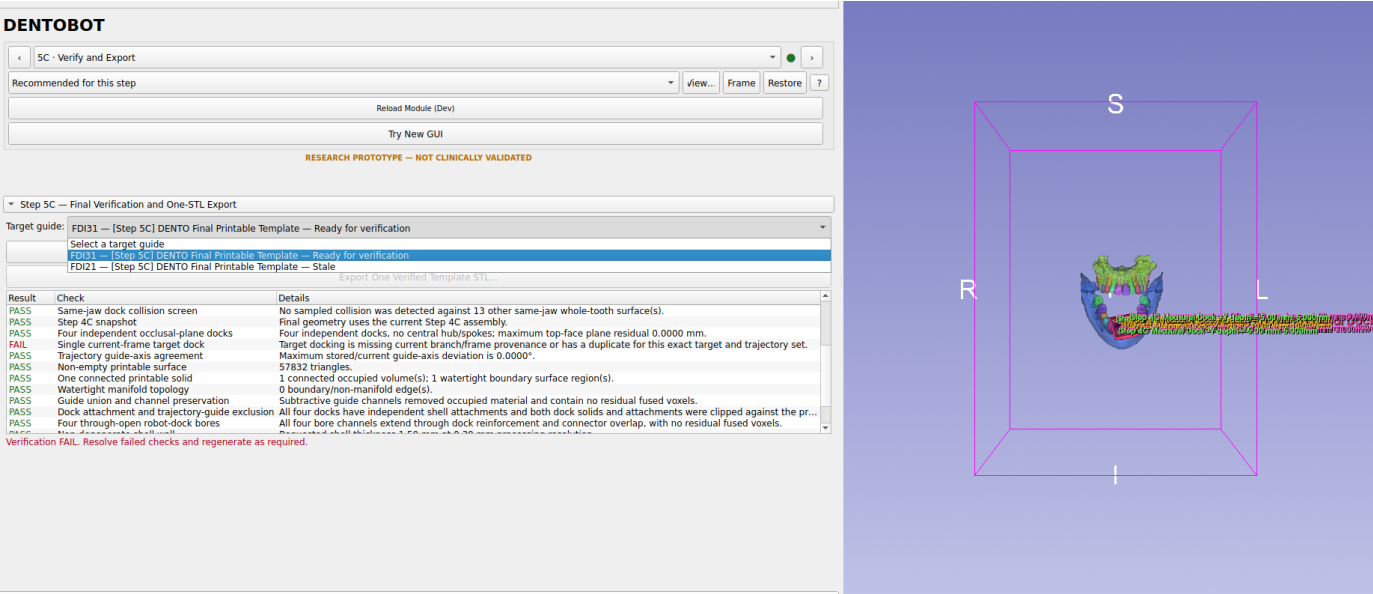


Failure text references exact target and trajectory set. Regeneration alone did not establish correct cross-target ownership.

Source: /tmp/codex-clipboard-c23214b9-a574-47eb-8479-45c23fe3eaea.png

Image SHA-256: da27a569f974e735ba80b448877f1c40d799de66d84f11b32818a692a165cb04

Figure 12 | Alternating guides exposes cross-target staleness

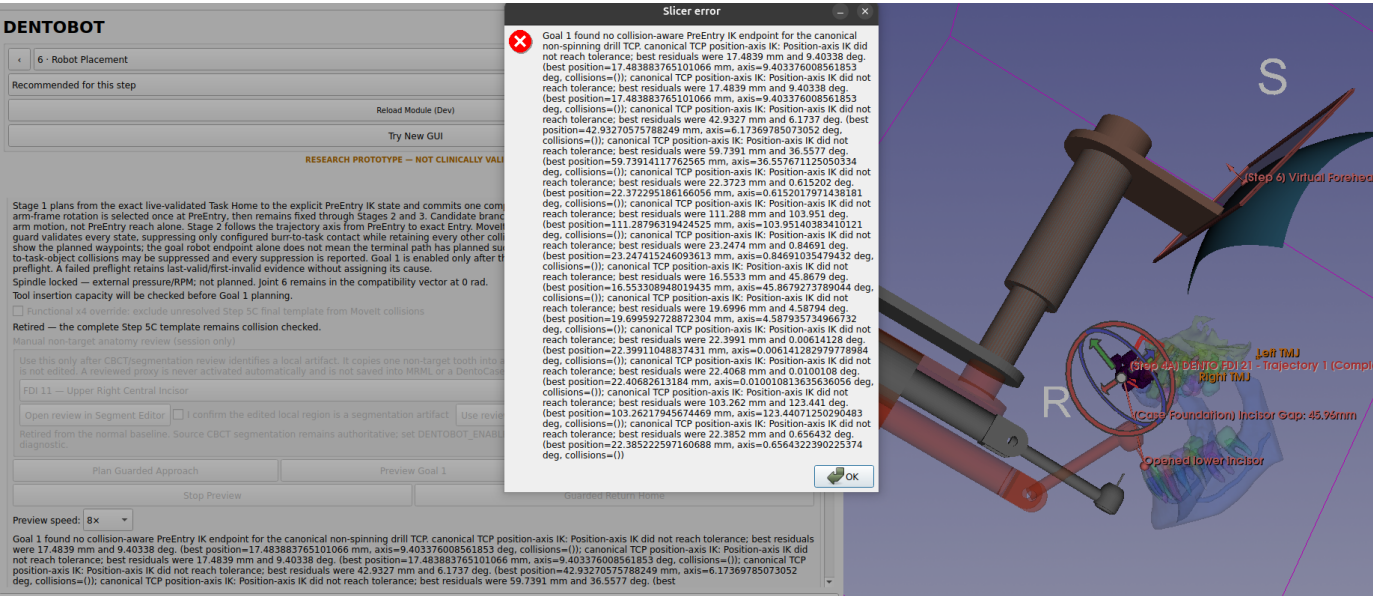


FDI31 Ready and FDI21 Stale in one selector. This is the coexistence problem the PreparedBranch work needed to address.

Source: /tmp/codex-clipboard-0e7602b5-2a77-4899-98fb-57db92b18d86.png

Image SHA-256: 9d7cd74dea111f1a2a29fc0e875ed96055718638bcc5e9ade22bc1d8349ea13c

Figure 13 | FDI21 planner reports no valid PreEntry endpoint

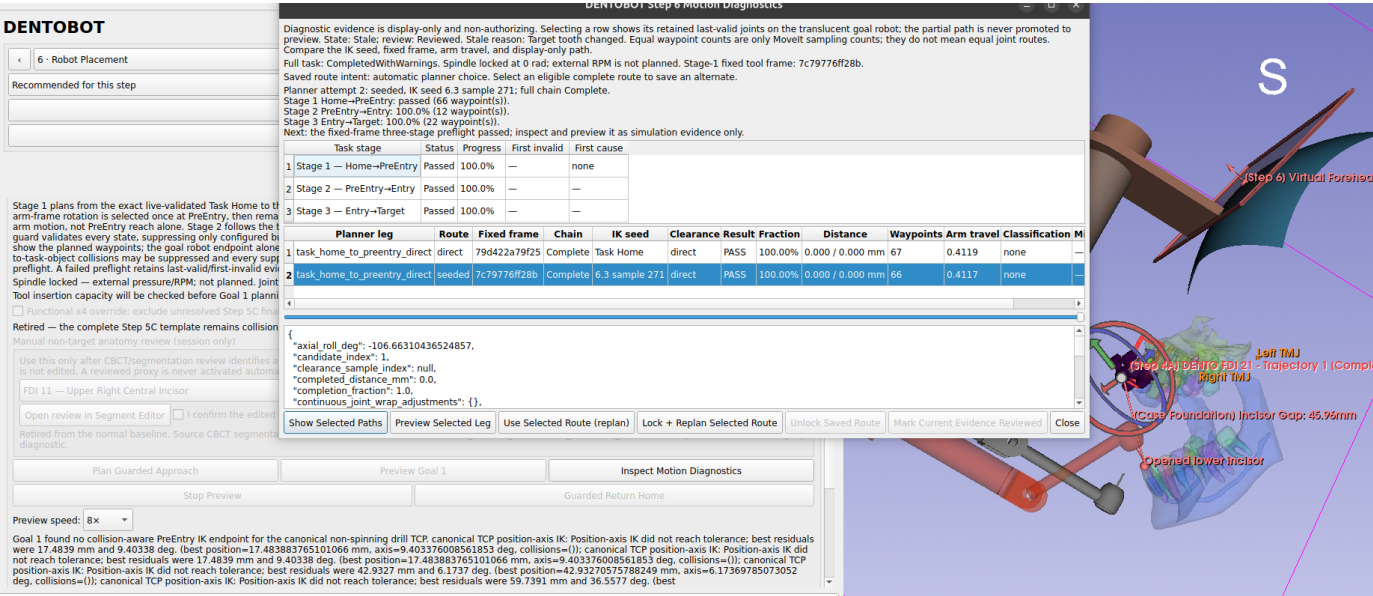


A fresh error shows position-axis IK residuals. It must not be combined with a stale complete-chain report to claim a current success.

Source: /tmp/codex-clipboard-1e7f43f7-05f9-495c-b3bc-6ab41a17f260.png

Image SHA-256: 540b53552e77ae58dada386148337306ce27a910cb4887c23bc825d05d159463

Figure 14 | Legacy complete diagnostics explicitly stale

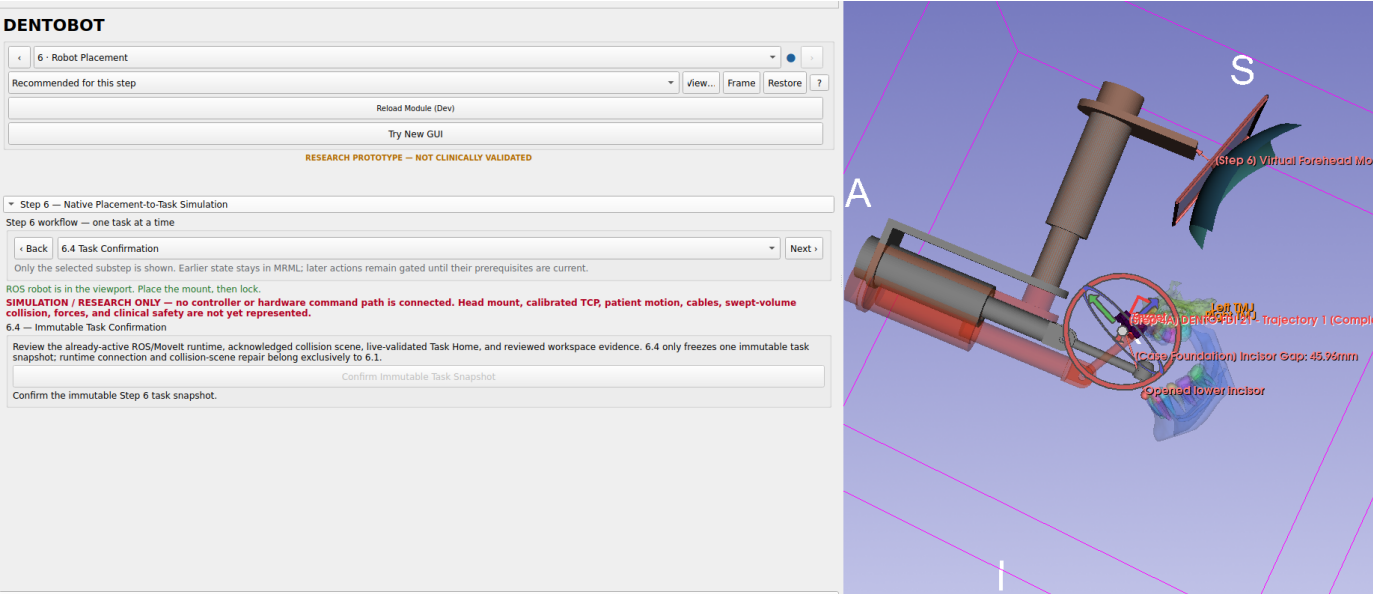


Top text: State Stale; reason Target tooth changed. The old 66/10/22 result is not the new FDI21 run.

Source: /tmp/codex-clipboard-f50d2446-be6c-461b-a05b-8a2eb8eacee6.png

Image SHA-256: 3cd4bdb94e81896704e4282363d016039d6de43641c8877ae073cbee212b4bb0

Figure 15 | Step 6.4 snapshot confirmation disabled

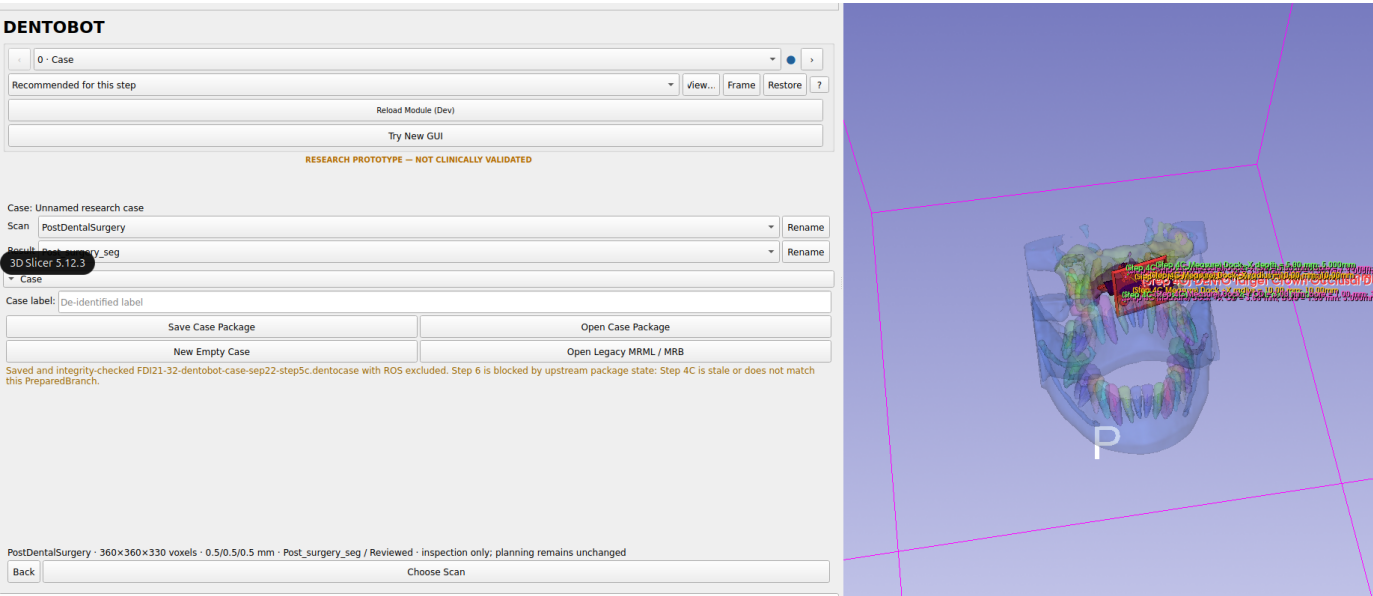


Operator had advanced through prerequisites but could not confirm. A disabled button requires prerequisite attribution, not a bypass.

Source: /tmp/codex-clipboard-1c95e5e1-42bd-4d7e-b0f4-dd43a382714c.png

Image SHA-256: 23788000f6c0ec623762c55a89352de625a60c71c480200ce4e5c4db85a5e3ff

Figure 16 | Case save preserves upstream branch mismatch

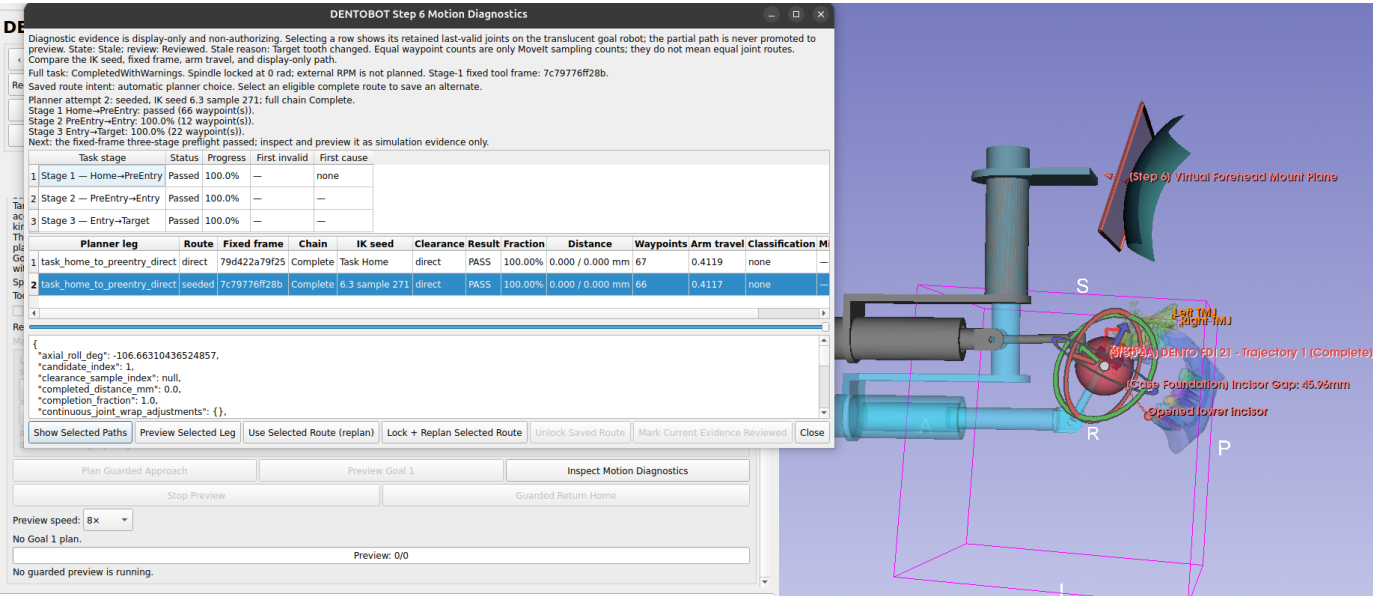


Saved and integrity-checked package explicitly reports Step 4C stale/mismatched to PreparedBranch. Archive integrity and workflow eligibility are distinct.

Source: /tmp/codex-clipboard-6d393a28-be84-4281-a320-4b0ce49b3ef2.png

Image SHA-256: c968499af9d7ab56abfdfb498c26327f905772ef4bdf68b885a45d561bf8e311

Figure 17 | Claimed FDI21 pass with stale evidence caveat

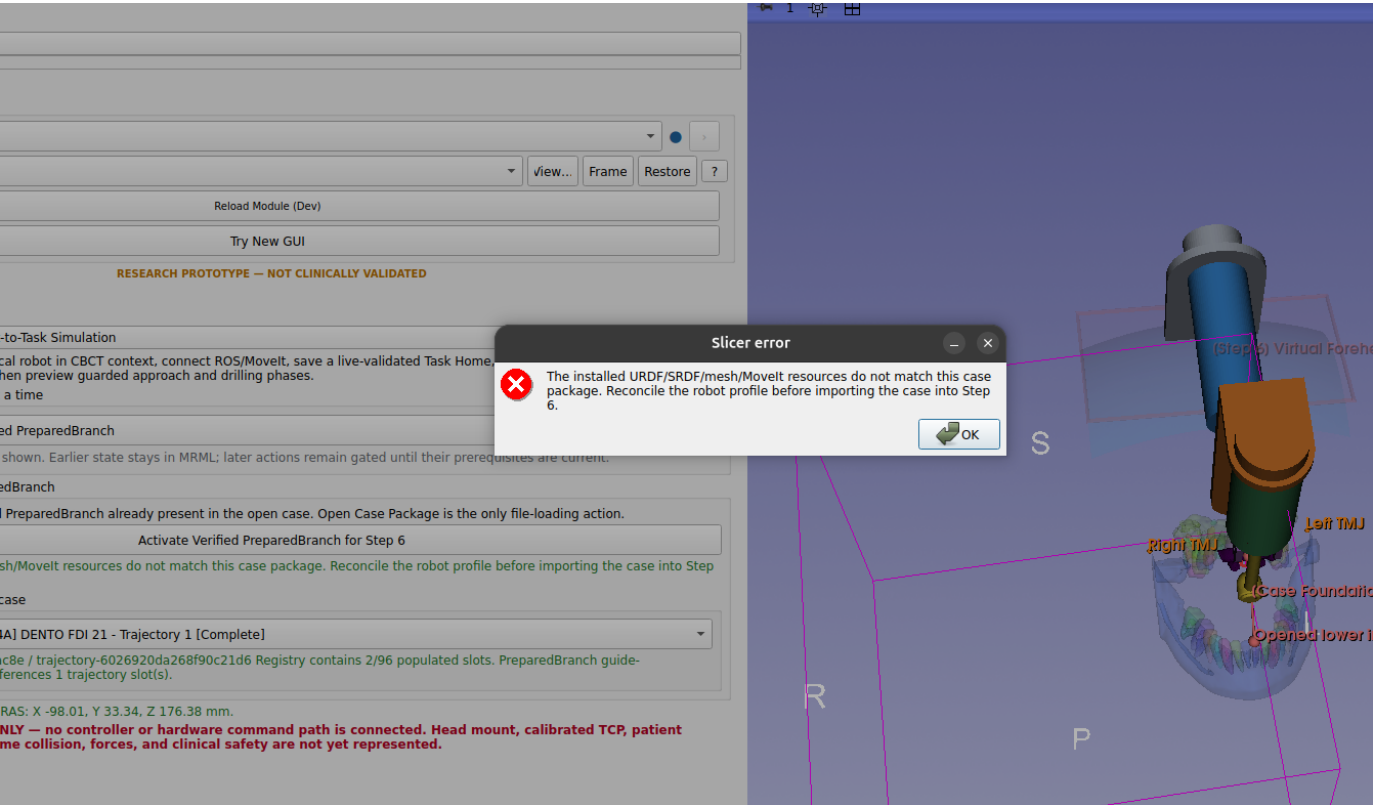


Operator reported manual base adjustment and success; displayed diagnostics still say target changed and underlying panel says No Goal 1 plan. Preserve the report, but this image alone does not certify fresh FDI21 full-chain acceptance.

Source: /tmp/codex-clipboard-c248b0dd-453c-415a-a74d-e27142688612.png

Image SHA-256: 5e1dd217b05e9d5302d712cfad3ccab97cedcc4a23bded7fdcbd31fc7017af6f

Figure 18 | Robot-profile compatibility gate

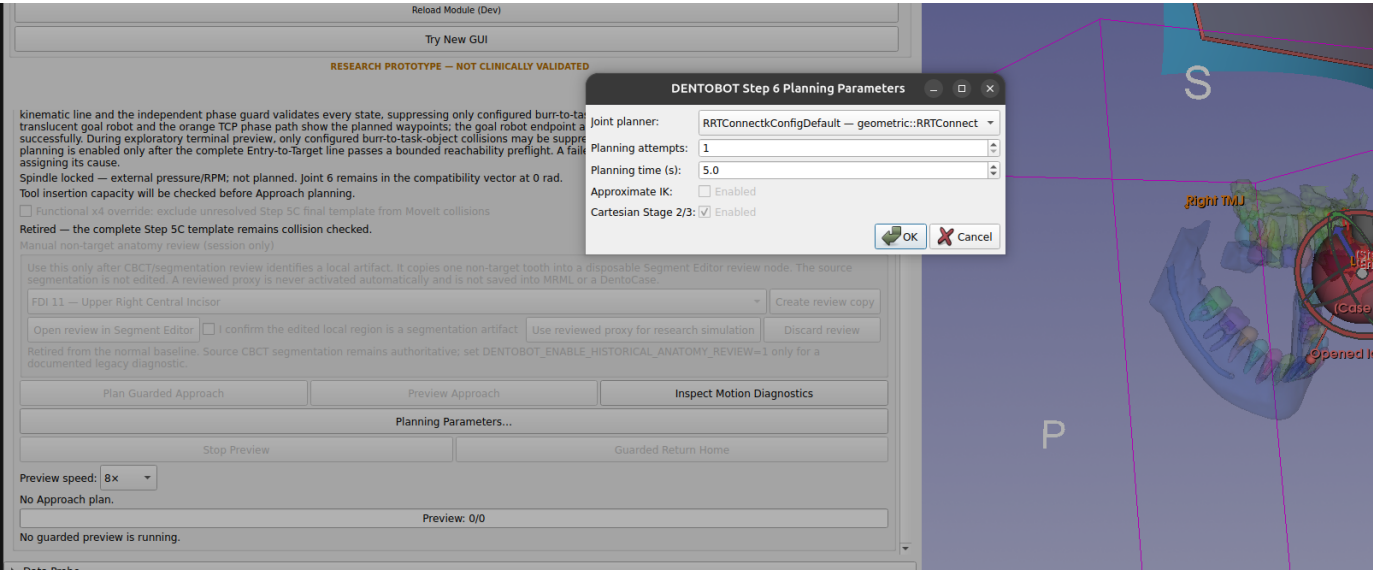


Installed URDF/SRDF/mesh/Movelt resources do not match the saved case. This prevents activation before planning and is not planner-performance evidence.

Source: /tmp/codex-clipboard-831a50d2-8a3f-42b5-bb73-04aae20251dc.png

Image SHA-256: 41935717fd8e5cc42d5ea1f10e4df1079058a1dcc6ae749f551c4f2f537cf5b2

Figure 19 | Planning controls and locked features

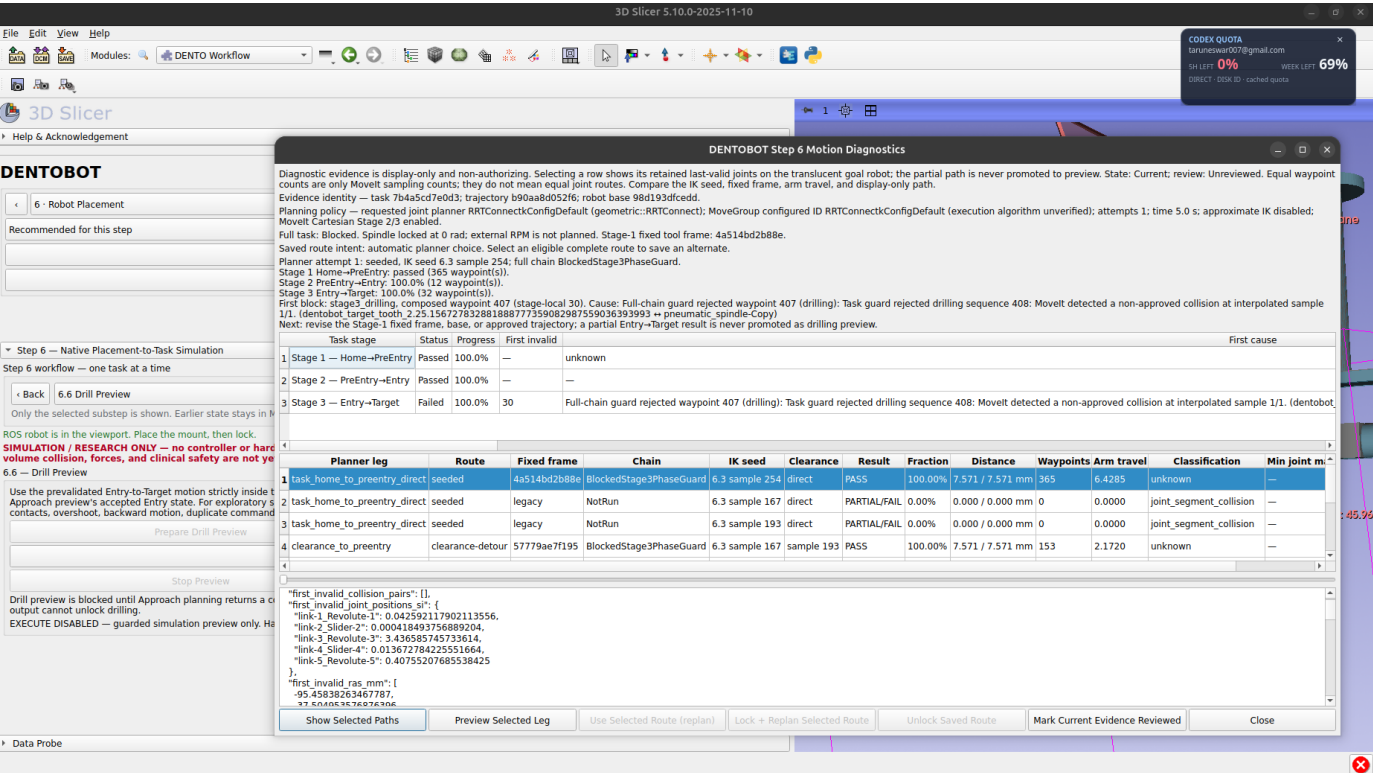


Initial shared parameters: RRTConnect, one attempt, 5 s; approximate IK disabled and Cartesian enabled but locked. Operator requested RRT*, tips and explanations.

Source: /tmp/codex-clipboard-07505912-ea0f-417d-9636-b3b115a86c5c.png

Image SHA-256: d02968bee7e4258591ed74f10bd68efa64efdd20cd56e4c9535a03fbcf35ff9f

Figure 20 | Configured planner and Stage-3 guard failure

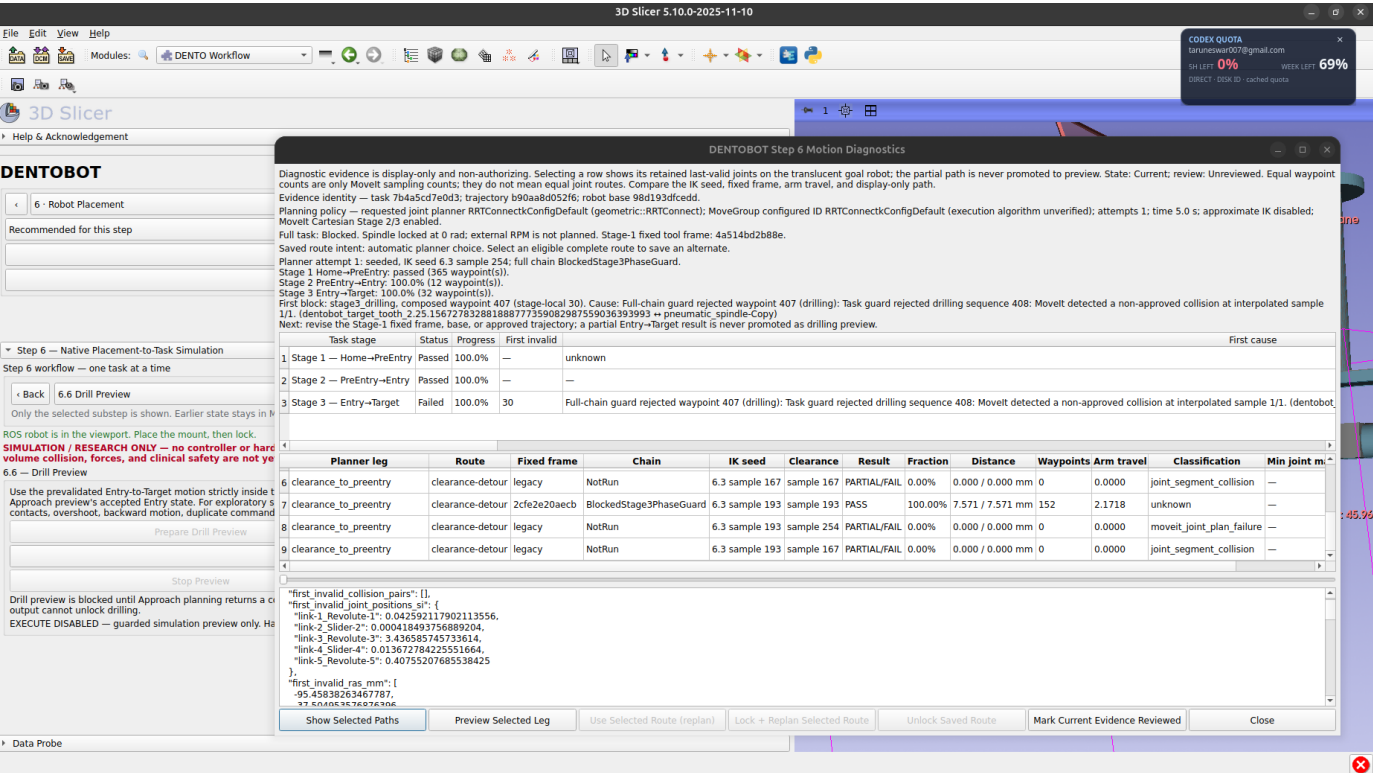


Current diagnostic shows configured RRTConnect, P1/P2 pass, Stage-3 non-approved tooth/spindle contact. Configured ID explicitly does not prove executed algorithm.

Source: /tmp/codex-clipboard-93734918-46ad-415f-bf2f-b5e9a01e3a52.png

Image SHA-256: 6f49b0dc3f633ccc42d56bd7793116b983af9afc5f6cdfd28223116758cd5c75

Figure 21 | Alternative candidate rows retain failures

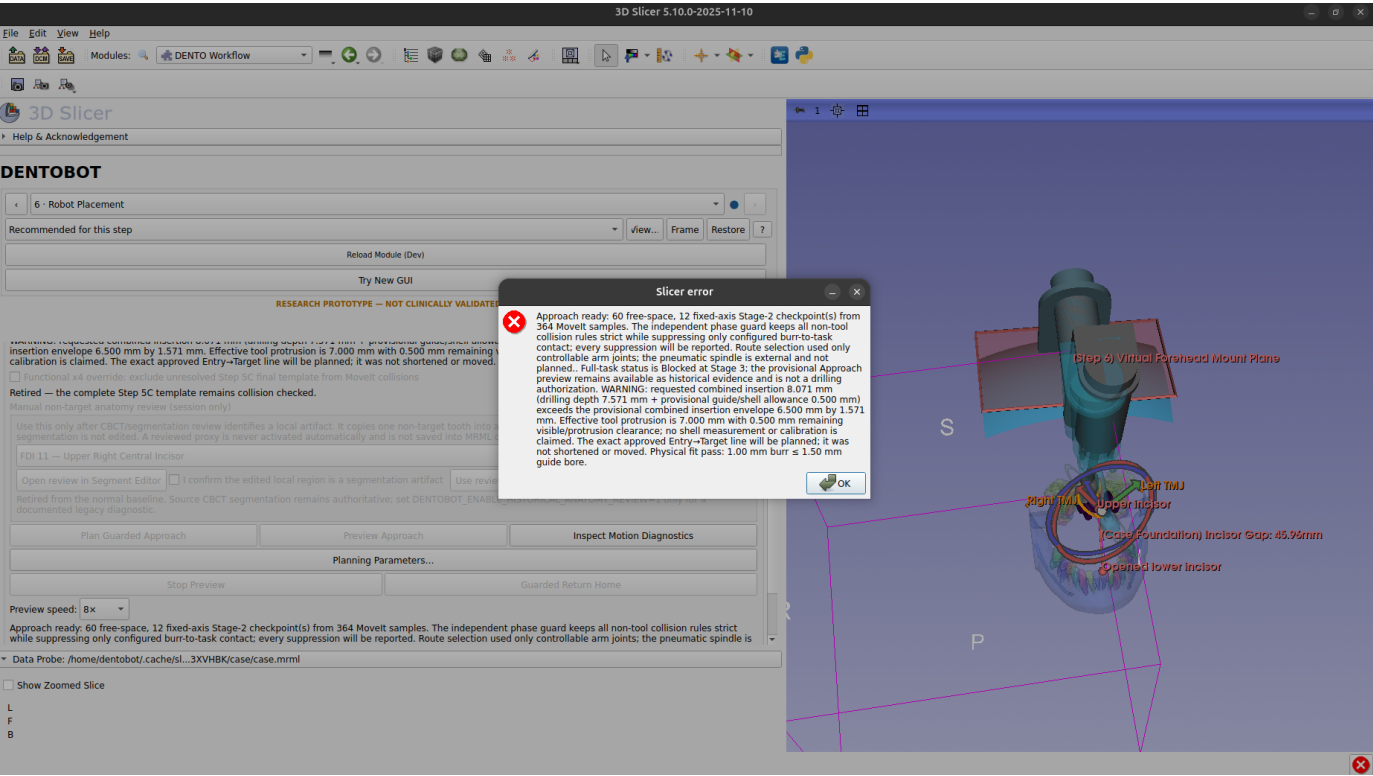


Direct and clearance candidate rows have mixed results and full-chain blocks. A route row PASS is not necessarily complete task acceptance.

Source: /tmp/codex-clipboard-d4bcb761-84c6-4e49-802c-546733908fd1.png

Image SHA-256: 09af702e2c910af4756bdb18aee64465239a9845d08024c864198574bb15f495

Figure 22 | Detailed result dialog and insertion warning

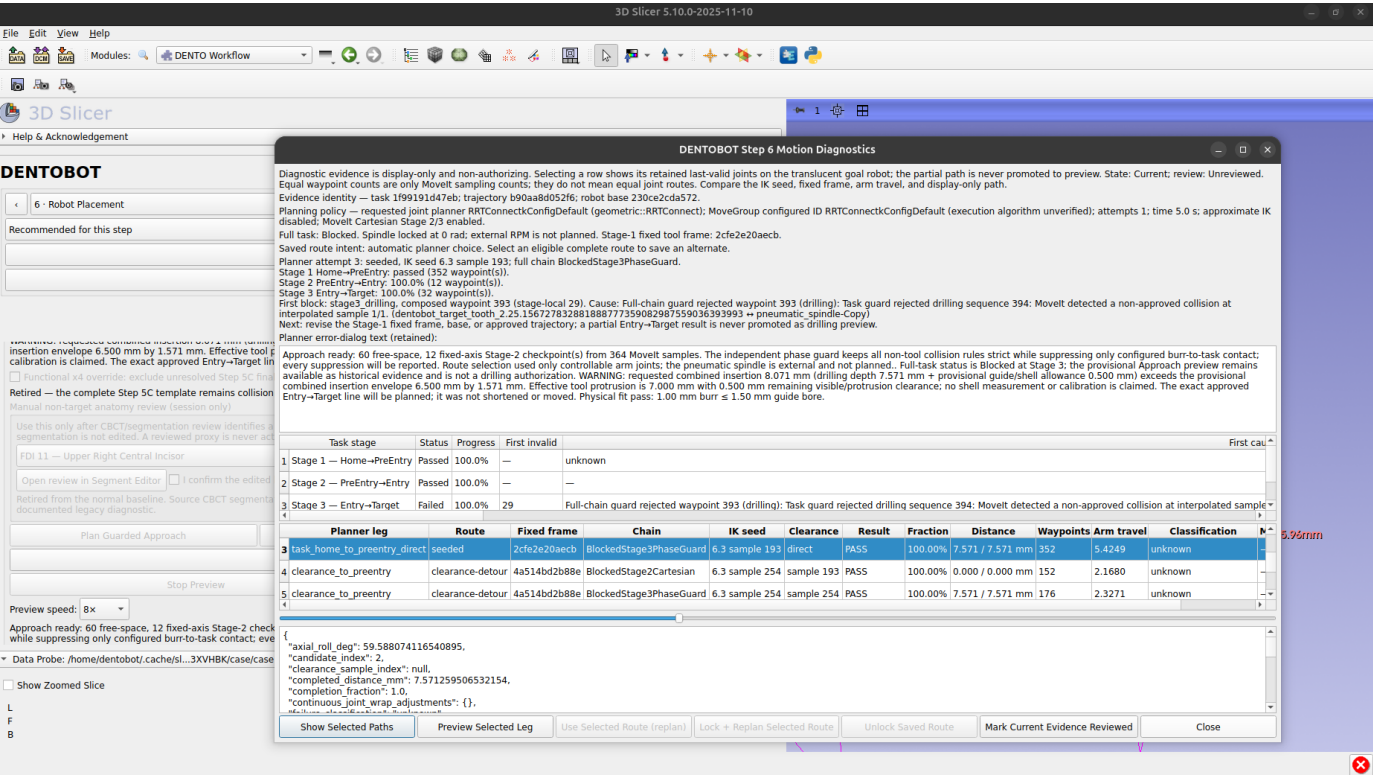


Approach ready while Stage 3 blocked; 8.071-mm combined insertion versus 6.5-mm provisional envelope. This message must remain recoverable after closing.

Source: /tmp/codex-clipboard-3bfda34a-24e7-4429-96cb-0a8824c0ee9d.png

Image SHA-256: 1926ca4fb585bdcff7439438767ca34c893b69b2750ac66c35e2892dd52554bf

Figure 23 | Operator-confirmed retained error text

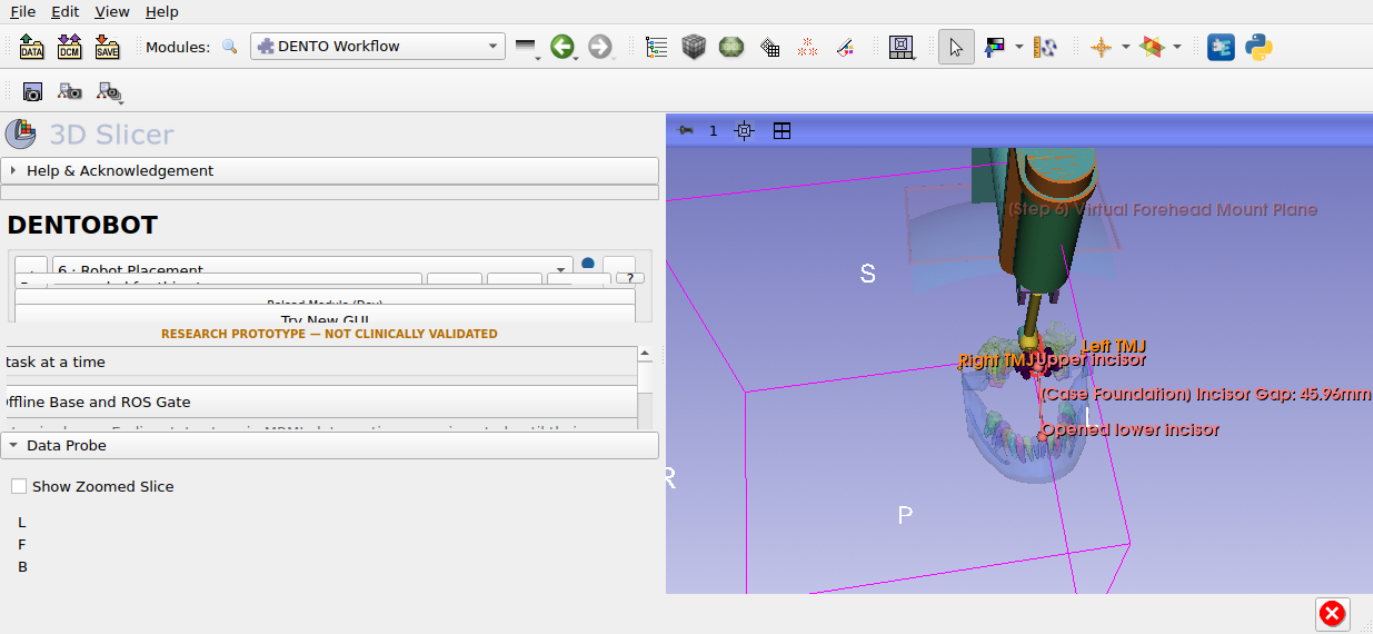


Motion Diagnostics includes the prior dialog text; user confirmed hover tips too. This is acceptance evidence for observability, not for the blocked drilling task.

Source: /tmp/codex-clipboard-30dcbdc4-c95e-48a2-a213-eca10f1bb055.png

Image SHA-256: c94728a52354577f7d39c11cecef7bd3a3b87f89c00f0755853fea0bcd12293c

Figure 24 | r6 automated base/runtime context

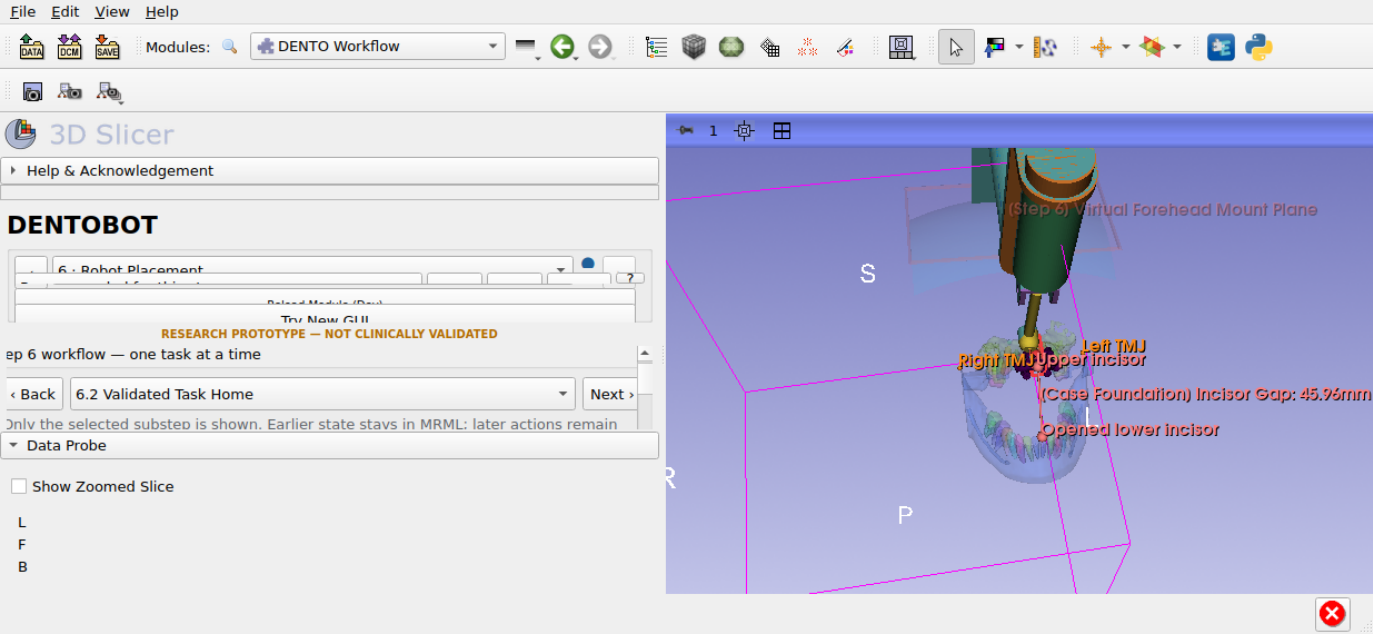


Retained setup image. View is tight; exact pose comes from numerical records, not visual estimation.

Source: /home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/FDI21-base-runtime.png

Image SHA-256: 2371b9158485df822f101720786685ab57cf2fa72607cb5b00fa317d0cf4ace2

Figure 25 | r6 automated Task Home evidence



Step 6.2 context retained separately after the earlier incorrect-substep capture. It does not substitute for the exact Home vector/fingerprint.

Source: /home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/FDI21-task-home.png

Image SHA-256: 6f2765a714743cc352ff03b5d677a29ad4ef07a37cdc4f1bda34d215e11359b3

Figure 26 | r6 completed RRTConnect checkpoint

Current case identity. Saved paths are display-only evidence, not ROS/MoveIt or guard-valid plans. Use a fresh single-planner replan before guarded preview.

	Planner	Result	Full task	First blocker	Waypoints
1	RRTConnect...	Fail	Blocked	Full-chain ...	461
2	RRTkConfig...	NotRun	NotRun		0
3	RRTstarkCon...	NotRun	NotRun		0

```
{
  "message": "Approach ready: 80 free-space, 12 fixed-axis Stage-2 checkpoint(s) from 429 MoveIt samples. The independent phase guard keeps all non-tool collision rules strict while suppressing only configured burr-to-task contact; every suppression will be reported. Route selection used only controllable arm joints; the pneumatic spindle is external and not planned. through previously validated workspace clearance sample 198. Full-task status is Blocked at Stage 3; the provisional Approach preview remains available as historical evidence and is not a drilling authorization. WARNING: requested combined insertion 8.071 mm (drilling depth 7.571 mm + provisional guide/shell allowance 0.500 mm) exceeds the provisional combined insertion envelope 6.500 mm by 1.571 mm. Effective tool protrusion is 7.000 mm with 0.500 mm remaining visible/protrusion clearance; no shell measurement or calibration is claimed. The exact approved Entry\u2192Target line will be planned; it was not shortened or moved. Physical fit pass: 1.00 mm burr \u2264 1.50 mm guide bore.",
  "paths": {
    "stage1": [
      {
        "link-1_Revolute-1": 0.0,
        "link-2_Slider-2": 0.0,
        "link-3_Revolute-3": 0.0
      }
    ]
  }
}
```

Replay Saved Path (display only)

Use Planner for Next Fresh Replan

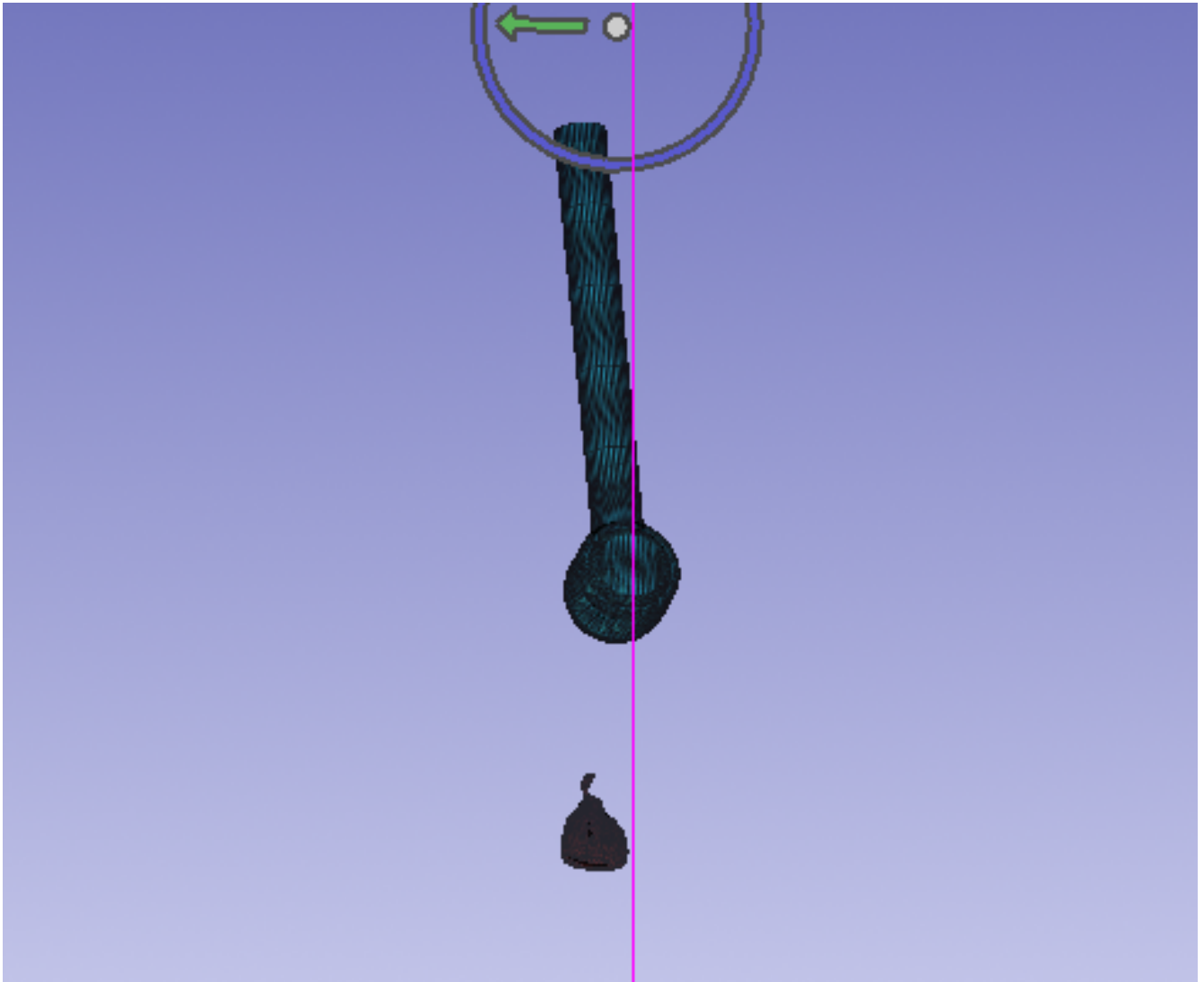
Close

Latest completed trial. Stage-3 collision persists; remaining planners do not have completed comparable outcomes.

Source: /home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/FDI21-rrtconnect-d
iagnostic.png

Image SHA-256: aa4591b3f09093cfaeeca22e90ec2ce2fc9604e2ff160b361aed82d678d33bf4

Figure 27 | INVALID contact evidence - inferior

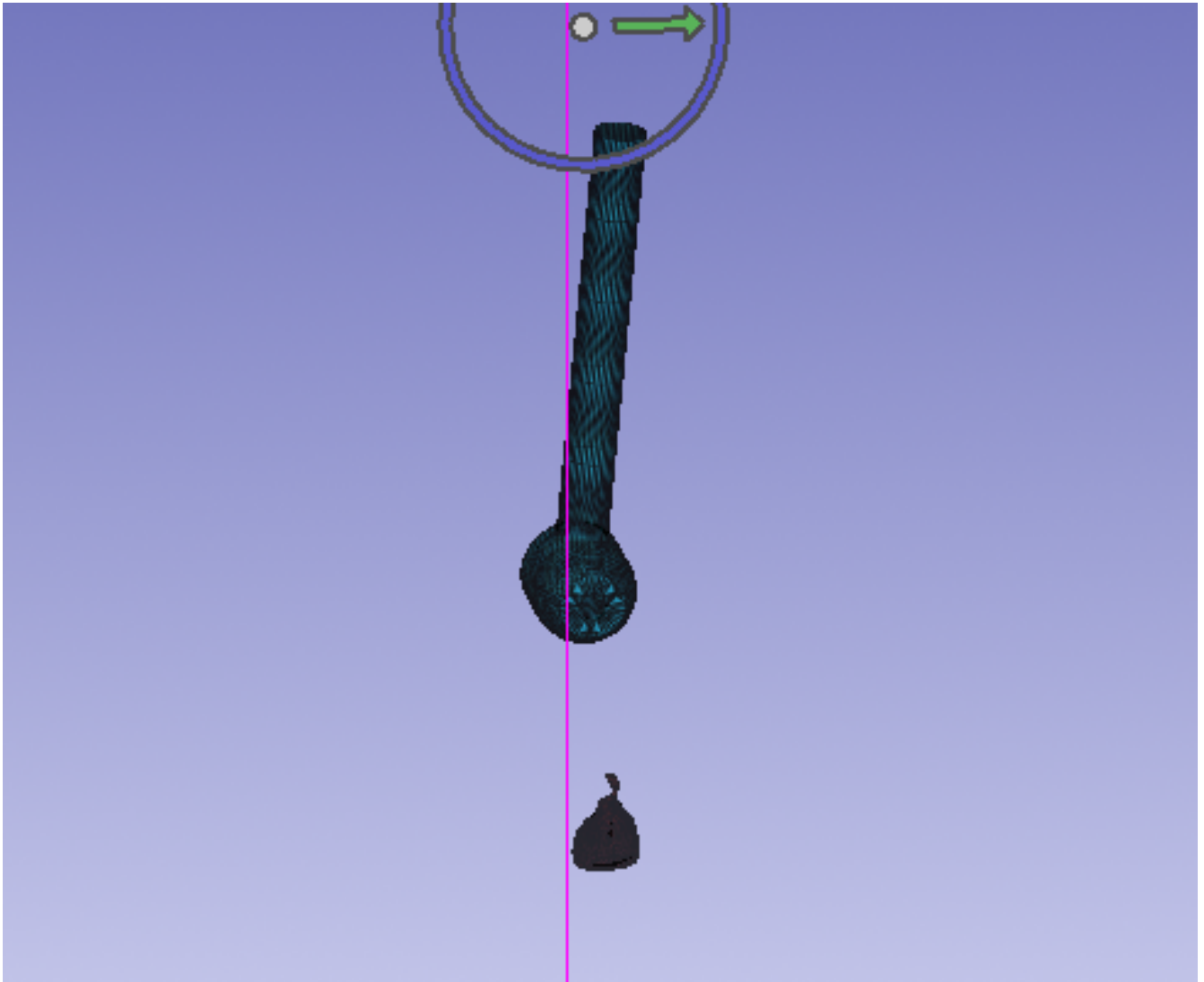


INVALID FOR CONTACT INTERPRETATION. Operator rejected this offline reconstruction. The script incorrectly applied the lower-jaw transform to fixed upper FDI21, producing separated model bounds. Included only to document the failed evidence path. It proves neither clearance nor native collision depth.

Source: /home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6-collision-recapture-r3/FDI21-first-invalid-inferior.png

Image SHA-256: b9647097449be695a8b692b3922a1dda45b1695903d4287c9fc4217ed1d1ec39

Figure 28 | INVALID contact evidence - apical

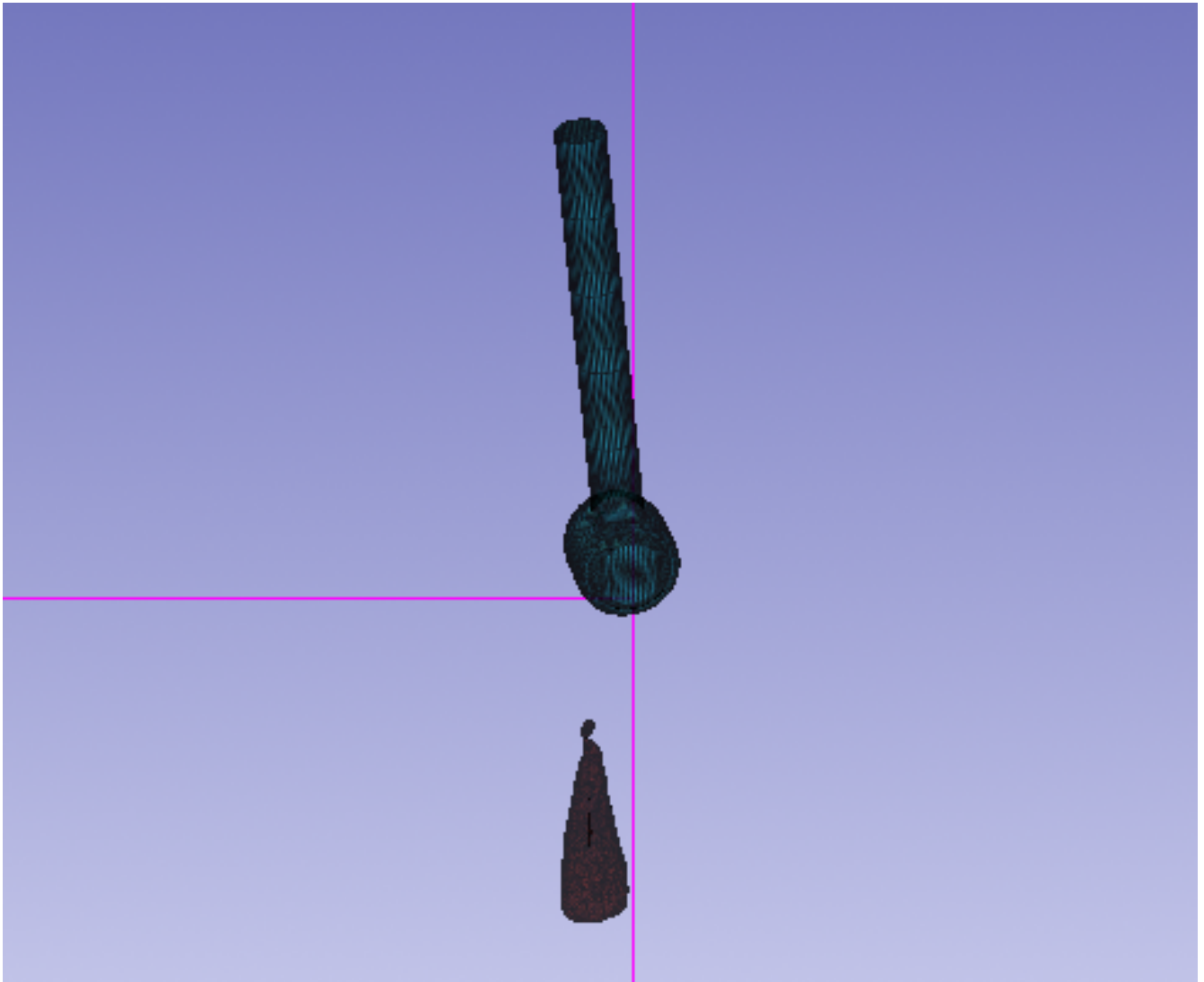


INVALID FOR CONTACT INTERPRETATION. Operator rejected this offline reconstruction. The script incorrectly applied the lower-jaw transform to fixed upper FDI21, producing separated model bounds. Included only to document the failed evidence path. It proves neither clearance nor native collision depth.

Source: /home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6-collision-recapture-r3/FDI21-first-invalid-apical.png

Image SHA-256: 9197fce56c44aca7426885b005d0a9533286ded4e6b6f67fce1037aecf9bb0d0

Figure 29 | INVALID contact evidence - oblique



INVALID FOR CONTACT INTERPRETATION. Operator rejected this offline reconstruction. The script incorrectly applied the lower-jaw transform to fixed upper FDI21, producing separated model bounds. Included only to document the failed evidence path. It proves neither clearance nor native collision depth.

Source: /home/light-tarun/dentobot/data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6-collision-recapture-r3/FDI21-first-invalid-oblique.png

Image SHA-256: 9e7ec373bbdc3e8b6790c5325f556ab850d8122e91f860b254773f6d09622aba

Appendix A. Detailed recorded execution chronology

Selected Sep-23 logbook excerpt, preserved as historical execution evidence. Later statements in this report supersede older “next” proposals in this excerpt. It is not a new execution checklist. Dates, outcomes and command claims belong to the original entries; they were not rerun during PDF preparation.

FDI32. The exact input is
data/Slicer_Saved/SampleStudy1/FDI21-31-headless-verified-sep22-step6a.dentocase
(SHA-256 c16e0406b589de1cb2144628b38a26100157458d48b5edb9c56ad4f8792d1c1d).
Read-only lineage inspection confirms an FDI21 branch
guide-fe8080de25e7eb57622d and an FDI31 branch
guide-5c0da27697aceceb8f14; FDI21 is selected at save. The preceding
FDI32 missing-input stop is resolved by this correction. Runtime execution
still awaits the protocol's explicit approval of the bounded serialized plan.

Runtime approval and attempt 1: Tarun approved the two-target serialized
simulation-only run. The first FDI21 headless Slicer process loaded the input
but failed before planning with Load and place the local MRML robot before
connecting ROS/MoveIt; startup also reported the ROS2 module unregistered.
Evidence: data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21/failure.json.
This was an automation setup failure, not a planner failure. No output case or
comparison screenshot was produced. The runner was corrected to call its
existing facade.loadRobot() step; the next bounded retry will use the
installed ROS2 module paths and the simulation stack handoff script. Failure
count for this setup blocker: 1.

Runtime attempt 2 and stop: The corrected handoff launched canonical
housing-on dentobot.urdf, reached simulation readiness, loaded the local
robot and synchronized the collision scene. It then stopped before planning:
validate saved Task Home: Task Home belongs to different robot resources.
Evidence:
data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r2/failure.json
and the handoff output/stack log. Read-only comparison shows the saved case
profiles moveit/config/ompl_planning.yaml at SHA
10f6f69a2f40f047b64430d1f408ecec0350ee29cafc27a9758d07821b16c355,
while the checked-in current YAML SHA is
9acb46e12a30dcf198d2fe415f10446b8f340fe57b9051de7a9ad91cbab5e7b3.
applyTaskHome() correctly rejects the stale robot-resource identity. The
stack handoff cleaned up. No planner trial, output DentoCase, or screenshot
was produced; FDI31 was not started. This is a new substantive profile/Home
blocker, not a planner outcome. The test will not auto-save a replacement
Home. Two runtime attempts total; further runtime is paused pending operator
direction on explicit current-profile Home recapture and case save.

Operator-approved current-profile recapture / attempt 3: Tarun explicitly
approved the recapture. The runner was restricted to the sole stale condition
Task Home belongs to different robot resources. and used production
saveTaskHome() (strict guard + monitored-state validation), regenerated and
reviewed workspace evidence, reconfirmed the task, and saved the non-overwriting
fdi21-r3/FDI21-current-profile-home.dentocase (SHA-256
d6da41b87c63549bdcffef06ce1df779f3ed396fe304ed826010147dfed0510c).
The first comparison call then tripped the batch's fatal identity check:
The case/task identity changed during planner execution. The saved
fdi21-r3/FDI21-three-planner.dentocase (SHA-256
c67a222c47d5bd23467d48780756ad40f69a345b93e6716cb2e25ca194b4168e)
contains the branch-keyed comparison with all three rows NotRun; the first
row carries that fatal text. It is not a three-planner result and must not
feed FDI31. fdi21-r3/result.json and FDI21-comparison.png were written,
but the screenshot captured the main window rather than the diagnostic dialog;
it is context evidence only. Slicer exited 1 with shutdown leak warnings after
functional artifacts were written; stack handoff cleaned up. The changed
identity field was not recorded by this runner and remains unknown. This is a

fatal comparison-integrity stop under S6-LIVE-01, not evidence that RRTConnect failed geometrically or that RRT/RRT* executed. No FDI31 run or fresh-process reopen followed. The three-attempt ceiling is reached for this run chain; pause for Tarun's instruction before any further runtime trial.

Follow-up source diagnosis and correction (no runtime): Tarun requested diagnosis and fixes. Read-only extraction of both saved MRML collision-audit records found exact prior/post fingerprints 778c1876...d6b6062 / 385c131a...1a3a6c741. Their 31 object records, base, jaw/world-to-base transforms, status and runtime acknowledgement match; only generated_at_utc differs (02:59:48.929178+00:00 versus 03:01:11.456064+00:00). The current audit_fingerprint includes that timestamp, and planApproachPhase() calls _prepare_phase_guard(), which resynchronizes/re-acknowledges the same scene. Thus the batch compared an ephemeral audit-event ID as if it were stable scene identity.

Fix: The comparison now freezes a separate content fingerprint of the acknowledged base/jaw/world-to-base/object manifest and guard policy. A changed object or policy still invalidates the batch; the exact per-trial audit ID is retained and checked against its diagnostic session. The headless screenshot helper now captures the visible comparison dialog rather than the main window. Changed DENTOSTep6State.py, DENTORobotWorkflowFacade.py, Testing/test_step6_state.py, and Testing/run_dentobot_step65_three_planner_gui.py. Focused pure tests: PYTHONPYCACHEPREFIX=/tmp/dentobot-pycache python3 -m pytest -q -p no:cacheprovider Testing/test_step6_state.py -k planner_comparison → 2 passed; changed Python compile and git diff --check passed. This is source/unit evidence only; no fourth Slicer/ROS trial was started. Next runtime plan requires explicit fresh approval under the retry ceiling. The full Testing/test_step6_state.py run also passed (28 passed). Graphify overlay refresh was attempted and again failed with [Errno 1] Operation not permitted; no updated graph is claimed.

Operator-approved attempt 4 (FDI21 only): Started the serialized simulation-only handoff from the recaptured fdi21-r3 Home case, with canonical housing-on robot and output under fdi21-r4/. result.json, FDI21-three-planner.dentocase (SHA-256 80828de0854d640cddd107d63e7ade2537a2888a3c6d0f3746dbd27f72cecbc9), and FDI21-comparison.png were saved. The first configured planner, RRTConnectkConfigDefault, returned a Stage-3 drilling guard failure after Approach/Stage 2 succeeded; the diagnostic identifies non-approved target-tooth ↔ pneumatic-spindle collision at interpolated sample 1/1. The RRTkConfigDefault and RRTstarkConfigDefault rows are not planner performance evidence: both failed before planning because Task Home was no longer validated in the current ROS/MoveIt session. This is not a valid three-planner comparison. No FDI31 run or offline reopen followed.

Diagnosis and source correction: planApproachPhase() checks the Task Home audit fingerprint, then _prepare_phase_guard() re-syncs the collision scene. That new, identical acknowledgement receives a new timestamp and audit ID; the next trial's Task Home exact-audit check consequently fails. In logic_robot_scene_sync.py, after a fresh scene acknowledgement the prior audit identity is now reused only if all audit fields except timestamp and fingerprint match exactly. Changed scene payload, policy, status, or acknowledgement still produces a new audit and invalidates Home. This is a source-only fix; no further runtime verdict is claimed. Commands/results: jq extraction of the r4 comparison; PYTHONDONTWRITEBYTECODE=1 python3 -m pytest -q Testing/test_step6_state.py → 28 passed; python3 -c compile of the changed sync module → pass. An initial py_compile was blocked by permission to write an existing __pycache__, so the non-writing compile check was used. Next stop: operator verdict/approval before another GUI trial; only a valid FDI21 comparison may feed the sequential FDI31 and offline reopen checks.

Operator approval and new evidence requirement: Tarun approved one corrected FDI21 runtime retry and required every diagnostic run to retain screenshot evidence and an explicit assessment of whether robot-base mount or Task Home changes could plausibly resolve a planner failure. This is a diagnostic question, not authorization to alter placement. Added

docs/diagnostics/STEP6_GUI_AUTOMATION_SOP_2026-09-23.md, linked it from the canonical plan and verification protocol, and updated Testing/run_dentobot_step65_three_planner_gui.py to capture a base/Home context image plus selected planner-row images and numeric base/Home state. Focused source checks: PYTHONDONTWRITEBYTECODE=1 python3 -m pytest -q -p no:cacheprovider Testing/test_step6_state.py → 28 passed; non-writing compile of runner and scene-sync module → pass; git diff --check → pass.

Approved FDI21 retry / attempt 5: Started the canonical housing-on stack with dentobot-simulation-slicer-handoff.bash, ROS domain 73, source fdi21-r3/FDI21-current-profile-home.dentocase, runner Testing/run_dentobot_step65_three_planner_gui.py, and fresh output data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r5/. Readiness passed; the case, robot and current Home/workspace gates loaded. FDI21-base-task-home.png was saved. The diagnostic process hit its 1200-s cap (exit 124); handoff reported cleanup_complete, and a read-only process check found no remaining Slicer/Movelt processes. No completed result JSON or DentoCase and no per-row screenshot were produced. Do not infer any full task outcome or use this run as the input for FDI31/offline reopen. Visual inspection of that image shows the robot/base context but the visible substep is 6.1 ROS Gate, not the Task Home controls: the script used substep index 1 instead of 2. The source-only correction now captures separate 6.1 base/runtime and 6.2 Task Home images; no corrected image is claimed for r5.

The retained fdi21-r5/stack.log contains planning-server messages for RRTConnectkConfigDefault → geometric::RRTConnect (25 configuration messages), RRTkConfigDefault → geometric::RRT (3), and RRTstarkConfigDefault → geometric::RRTstar (10). This is stronger algorithm configuration evidence than the earlier Slicer getter, but it does not provide complete per-trial guard outcomes. Those counts are internal planning calls, not three complete trial counts or a fair speed ranking. The long batch spans many Movelt/Cartesian/guard calls despite a 5-s allowance per individual native plan request. The precise active call at timeout remains unknown.

Source-only runner correction after timeout: The script previously waited until the final dialog to capture every row. It now intercepts its own comparison-progress save in the GUI process, writing checkpoint.json and a selected-row screenshot immediately after each completed trial. This changes test evidence collection only, not production planning/guard logic; it has not received a runtime check. The fdi21-r5 timeout is a new execution-cost blocker; no automatic retry or FDI31 continuation follows. Base/Home assessment: the earlier FDI21 Stage-3 target-tooth↔spindle collision may depend on whole-chain posture, so mount placement could matter; changing Home alone can affect Stage-1 route/IK seed but cannot be credited with resolving a Stage-3 target contact without a controlled same-target test. The timeout provides no additional geometric verdict. Next bounded action requires Tarun's review of cost/evidence and explicit approval for another runtime plan; keep base and Home fixed during any three-planner comparison.

Final source-only checks after the checkpoint and substep corrections: non-writing compile of Testing/run_dentobot_step65_three_planner_gui.py and git diff --check passed. Workspace/scripts/graphify-update.bash was attempted and failed with [Errno 1] Operation not permitted; no refreshed graph is claimed. Changed/added files in this continuation: runner, logic_robot_scene_sync.py, docs/diagnostics/STEP6_GUI_AUTOMATION_SOP_2026-09-23.md, canonical plan, verification protocol, backlog, decisions and this logbook.

Operator timing question and SOP delta: Tarun requested three fix options for the FDI21 timeout and emphasized timestamped automation diagnostics. A read-only stack-log split shows first planner-configuration evidence at epoch 1790162597 (RRTConnect), 1790163053 (RRT), and 1790163434 (RRT*); cleanup began about 1790163647. Thus the respective configured-planner spans were approximately 456 s, 381 s, and at least 213 s, after roughly 150 s of setup. These spans include candidate search, repeated Movelt calls, Cartesian preflight, scene/guard work and reporting; they are not native solve durations or a performance ranking. Source confirms planApproachPhase() may examine up to 14 IK seeds and then bounded workspace-clearance detours, each potentially issuing its own native joint plan and downstream chain preflight. The 5.0-s

setting bounds an individual native planning request, not the entire trial. The automation SOP now requires flushed UTC and monotonic timestamps at setup, trial/phase boundaries, screenshot/save and timeout so a future cap leaves the active phase identifiable. This is documentation only; the runner does not yet implement that timing instrumentation.

Operator choice — A now, C after evidence; total-budget option B rejected: Tarun explicitly eliminated the proposed total per-planner trial budget and requested timestamp instrumentation, then measured optimization only after the first output. This does not rescind the separate earlier Task Home Experiment B in the canonical planner plan. Tarun also objected to treating base/Home reasoning as an afterthought. The r4 FDI21 diagnostic is specific: Home→PreEntry and PreEntry→Entry passed, while Stage 3 hit a target-tooth→pneumatic-spindle guard collision. A different Home could change selected candidate/IK seed; a different mount could change the available whole-arm/tool-frame options. Neither directly removes a collision at the identical TCP/tool frame, and the r5 timeout adds no new geometry verdict. These are hypotheses requiring frozen-case comparison, not automatic changes.

A source implementation: Testing/run_dentobot_step65_three_planner_gui.py now appends and fsyncs timeline.jsonl events with UTC, monotonic elapsed, planner and call IDs, contexts, durations and failure boundaries. Test-only wrappers time the existing Goal-1 trial, scene/guard setup, PreEntry IK, clearance-candidate generation, joint plans, joint-segment diagnostics, Cartesian plans, chain preflight and waypoint guard. The already added per-trial checkpoint.json now contains a conservative first-failed-stage base/Home hypothesis, along with frozen placement evidence and screenshots. No production planner/guard method, candidate limit, robot pose or collision policy was changed by this A instrumentation. Testing/test_step6_state.py contains one focused pure check for UTC/monotonic timeline output and the Stage-3 hypothesis boundary. Commands: Graphify query of the facade/callback path; source and r4/r5 log inspection; focused pytest
PYTHONDONTWRITEBYTECODE=1 python3 -m pytest -q -p no:cacheprovider Testing/test_step6_state.py -k step65_runner_timeline_and_placement_hypothesis → 1 passed; non-writing compile of the runner → pass. No Slicer/ROS runtime was started. Next: inspect the exact A diff and run git diff --check; then seek a new bounded runtime approval. Option C is limited to the measured dominant repeated work after A evidence, with fresh guard checks retained. Final focused check includes the timing wrapper itself: the same test passed after asserting paired phase_start/phase_end events and nonnegative duration (1 passed, 28 deselected). git diff --check and non-writing runner compile passed. Graphify overlay refresh was attempted and again failed with [Errno 1] Operation not permitted; no graph freshness claim is made.

Operator-approved A runtime, FDI21 r6 (one bounded attempt): Tarun approved one 20-minute, simulation-only, canonical housing-on comparison from the recaptured current-profile Home case; no base/Home change, hardware, FDI31 continuation or retry was authorized. The source SHA-256 was d6da41b87c63549bdcffef06ce1df779f3ed396fe304ed826010147dfed0510c. Preflight focused pytest passed (1 passed, 28 deselected), git status confirmed the existing dirty branch, and the container process check found no active Slicer/Movelt stack. Command: docker exec dentobot-slicerros2 bash -lc with ROS Jazzy/install sourced, ROS_DOMAIN_ID=73, DENTOBOT_DIAGNOSTIC_NO_SPINDLE_COLLISION=false, FDI21 source and r6 evidence paths, then dentobot-simulation-slicer-handoff.bash --stack-log ... -- timeout 1200s xvfb-run -a ros2 launch slicer_ros2_module slicer.launch.py using Testing/run_dentobot_step65_three_planner_gui.py. Stack startup explicitly reported dentobot.urdf, diagnostic housing-off false. Exit 124 at the cap; handoff reported cleanup_complete.

Evidence directory:
data/dentobot-runs/step65-three-planner-fdi21-31-20260923/fdi21-r6/.
timeline.jsonl records case open at 60.4 s, Step 5C verified at 66.5 s, PreparedBranch activated at 70.5 s, ROS connected at 107.0 s, and Task Home validated at 109.5 s. It contains phase timestamps and the complete RRTConnect checkpoint at 1050.4 s. checkpoint.json freezes branch, task, trajectory, profile, base, Home and collision-audit identities; it marks RRTConnect Fail, RRT and RRT* NotRun at checkpoint time. The RRTConnect

trial spent approximately 874 s; eight completed guard_waypoints calls accounted for 813.39 s (largest 141.8 s), while 14 joint-plan calls totaled 4.6 s and 14 Cartesian calls 21.97 s. These are nested phase durations, not independent totals to add. RRT began, but timeout interrupted its second waypoint guard; no completed RRT or RRT* result exists and no combined result.json or saved comparison DentoCase was produced.

The completed RRTConnect outcome passed Approach via workspace clearance sample 198, then full-task Stage 3 was blocked at composed waypoint 459: non-approved dentobot_target_tooth_2.25.156727832881888777359082987559036393993 ↔ pneumatic_spindle-Copy collision. It also reports requested combined insertion 8.071 mm exceeding provisional 6.500 mm envelope by 1.571 mm. This is an operator-observed/agent-extracted diagnostic outcome, not accepted planner success or drilling authorization. FDI21-base-runtime.png, FDI21-task-home.png, and FDI21-rrtconnect-diagnostic.png are retained. The base/Home images share a tight view and show 6.1/6.2 state rather than quantitative coordinates; exact base/Home fingerprints are in checkpoint. The placement review remains a hypothesis: changing Home can alter candidate selection, and base placement can expose another reachable tool frame, but neither directly removes a collision at the same TCP pose. No pose was moved. Next stop: Tarun's explicit verdict on this GUI evidence; then a separately approved, bounded option-C optimization of repeated full guard work before another three-planner runtime attempt. Do not silently raise the cap or infer relative planner performance from this incomplete comparison.

Operator correction after r6: Tarun states that collision-caused planner failures need extensive, multi-angle screenshot evidence from the already written test-script pattern before asking for his advice. A table/text-only headless result deprives him of the geometric context needed for expert intervention and invites an agentic fix/approval loop. He notes this three-stage planner has been under work for over 21 days. This is his stated observation and process requirement, not an agent-inferred geometry verdict.

Source audit and bounded response: Backlog lookup routed to S6-LIVE-01; Graphify located existing diagnostic capture scripts. Read the current three-planner runner, Step 6 GUI SOP, prior FDI21 three-angle capture in Testing/run_dentobot_step65_exact_case_smoke.py, comparison callback and retained r6 checkpoint. The checkpoint has the selected candidate's first-invalid J1-J5 vector and target-tooth/spindle IDs, but r6 has no first-invalid scene image. Its diagnostic-table screenshot cannot be recast as one. Changed the runner to reconstruct the selected first-invalid pose on the display-only goal robot, isolate the collision-audit target and goal spindle, and capture inferior, apical, and oblique views. It records exact joint state, pair, paths, or an explicit unavailable reason in the per-trial checkpoint; missing collision views cancel remaining trials for operator review. It does not change native collision geometry or authoritative robot state. Updated the SOP and canonical contract with the mandatory visual handoff; kept it under the existing task, not a parallel work item.

Focused source check:

```
PYTHONDONTWRITEBYTECODE=1 python3 -m pytest -q -p no:cacheprovider
Testing/test_step6_state.py -k step65_runner_timeline_and_placement_hypothesis
→ 1 passed, 28 deselected. No Slicer/ROS/runtime was started for this source
change. Next: inspect diff, run git diff --check, then stop for Tarun's
manual review; a new collision-image runtime trial is not implied by this
source-only change.
```

Final non-writing runner compile and git diff --check passed. The required Graphify overlay refresh was attempted and failed with [Errno 1] Operation not permitted (exit 1); no refreshed graph is claimed. Changed files for this correction: Testing/run_dentobot_step65_three_planner_gui.py, Testing/test_step6_state.py, the Step 6 GUI SOP, canonical plan, backlog and this logbook. No new runtime screenshots exist yet; r6 remains text/table-only collision evidence and cannot be offered as the requested visual verdict.

Operator-approved offline FDI21 visual recapture: Tarun explicitly approved the bounded display-only recapture proposed after r6, not another planner run. Added Testing/run_dentobot_step65_collision_recapture.py to open the SHA-256 verified fdi21-r3/FDI21-current-profile-home.dentocase, read r6's retained RRTConnect selected candidate 6/first-invalid J1-J5 state, compare the saved

base fingerprint, rebuild local URDF robot meshes via `createOrUpdateRobotPlacement` and `updateRobotJointPoses`, prepare the selected target surface using the existing collision-scene jaw transform, and render inferior/apical/oblique views with no ROS connection, planner call, base/Home edit or scene publication. Non-writing compile and git diff --check passed before execution. The exact command used docker exec dentobot-slicerros2, sourced ROS environment for the installed Slicer launcher only, and invoked timeout 180s xvfb-run -a ros2 launch slicer_ros2_module slicer.launch.py with this Python script and a new output directory each time. No move_group/guard stack was launched.

Attempt 1 (fdi21-r6-collision-recapture/) failed before images because this Slicer build lacks `qMRMLThreeDView.resetCameraClippingRange`; failure.json records the exact exception. Attempt 2 (...-r2/) used the renderer clipping method and saved a manifest plus three PNGs, but image inspection showed only the tooth, not the spindle. Attempt 3 (...-r3/) framed both saved-model bounds and saved three PNGs plus manifest.json. The manifest records case SHA d6da41b87c63549bdcffef06ce1df779f3ed396fe304ed826010147dfed0510c, checkpoint SHA 49115ceaf8d3cfcf8e72444a81d6ca74e694d41587dfe97e9805be3fdc7883b1, base fingerprint 230ce2cda5723fbc41fadfd9c1f0f887710b29a444cbf73bee4daddaa394e47c, the exact five joint values and the native tooth↔spindle pair. The target model world-RAS bounds are [-99.511,-91.399,-65.894,-54.356,18.623,41.046] mm; spindle visual-model bounds are [-106.345,-85.794,-40.817,22.878,38.431,75.941] mm. Their Y bounds have a 13.54-mm axis-aligned gap in this reconstruction. Visual inspection likewise shows a separated spindle and tooth in all three angles, contradicting the native guard's reported collision at waypoint 459. The visual mesh and collision mesh share the same canonical URDF mesh/origin, but this does not establish that our offline scene reconstructs the native planning scene correctly. Treat the mismatch as unresolved state/transform/attribution evidence, not proof of native clearance or authorization to change collision policy. The launcher returned exit 0 while logging Slicer child exit 1 after the manifest was written; this shutdown discrepancy is also retained. No Slicer/Movelt processes remained afterward. Three scoped capture attempts have reached the retry ceiling; no fourth run or planner retry. Next bounded step is Tarun's review of the three r3 images and the mismatch packet, then a source-only native-vs-display transform attribution audit if he directs it.

Operator rejection of recapture evidence: Tarun explicitly rejected the three r3 screenshots as invalid collision evidence: they show an aerotor spindle and a tooth-like object but do not show the colliding object in contact, so they do not support expert geometry advice. This supersedes the preceding proposed request for his visual verdict. The images remain archived only as evidence of a native/display reconstruction mismatch. No further runtime or planner trial was started. Source audit traced the retained `first_invalid_joint_positions_si` to the waypoint rejected by `validate_task_phase_waypoints`; it is a waypoint-level vector while the native message reports collision at an interpolated sample. The target model was recreated from saved segmentation plus jaw preparation, and the spindle from local URDF FK rather than the live native planning scene. The exact cause of their 13.54-mm Y separation remains unresolved; do not claim it is solely interpolation, transformation, or a false native report. A minimal source-only fail-closed bounds-overlap check now prevents both the offline recapture and future comparison runner from calling separated model images collision views. Next: attribute the native scene object transform and exact rejected sample against the displayed models source-first, then capture only if equivalent state/geometry can be shown. Do not ask Tarun to diagnose from the invalid r3 screenshots or seek routine approval for another blind trial.

Appendix B. Final interpretation and report provenance

Bottom line: there is a real recorded native guard rejection, a separately identified diagnostic-image defect, an incomplete comparison, and a measured performance bottleneck. Fixing one must not be presented as fixing all four.

Path A makes sense if it is run as finite evidence gates. The immediate implementation is diagnostic scene/state attribution, followed by one bounded exact-state check if approved. Base/Home reasoning is integral to the next causal decision, but only after the geometry being reviewed is the geometry actually evaluated. A manual tune-until-pass strategy is not the recommended foundation.

Report production used read-only Graphify navigation, controlled-document/source inspection, retained JSON/log and image files, Python ReportLab layout and PDF rendering. It did not start Slicer, ROS, MoveIt, load a saved case into runtime, move a robot, alter anatomy, run a planner, change guard policy or accept a milestone. Existing source edits and input cases were preserved.

Prepared 23 September 2026. The latest operator request is for time and evidence to make an informed decision. No further runtime action is implied by delivery of this report.